

Group Optimization Report

Objective (aggregated fitness): 0.0227826

Penalty (total aggregated): 0

Algorithm: Classic + Nelder-Mead (final polish)

Population size: 384

Parameter vector: 32 params (shared[0-10], Bas_2[11-17], Bas_1[18-24], Bas_0[25-31])

Input file: propellants.01234.json

Generated: 2026-08-03 09:41

Run Configuration (as resolved)

Provenance

Mode: optimization

Configuration source: /root/runs/6h-Classic-Tm2300-kc400-constrON-fsEquilibrium-20260803-0118/run-config.json

Working directory: /root/runs/6h-Classic-Tm2300-kc400-constrON-fsEquilibrium-20260803-0118

Input propellants file: propellants.01234.json

Resolved sidecar: run_configuration.resolved.json

Recorded at (UTC): 2026-08-03 03:40:15

Model constants

Metal melting temperature, K: 2300

Skeleton conduction contact factor: 400 (contact-area fraction 0.0025; fixed calibration, not fitted)

Skeleton surface fraction: equilibrium carbon: $f_s = n_C(s)(T_{pore}, p) / n_{C,total}$, $T_{pore} = (T_{flame,skeleton} + T_s)/2$ — no fitted coefficients, no agglomeration data

coverage table: skeleton_carbon_equilibrium.json

Search

Strategy: Classic

Parameter vector dimension: 32

Population size (effective): 384

Worker processors (effective): 32 of 36 machine cores

RNG seed: 20260729 (reproducible at 32 workers)

Termination: stagnation streak 2,147,483,647 @ rel 1E-06 OR 6 h safety timeout

Mutation force F: 0.5

Crossover probability CR: 0.9

Nelder-Mead refinement

Enabled: yes

Stages: final polish (100,000 evals)

Adaptive coefficients: yes

Domain tolerance: 1E-08

Function tolerance: 1E-08

Restarts: 2

Penalty thresholds

Penalty rate: 0.01

Pocket heat-flux ratio threshold: 100

Pore diameter threshold: 3

Large oxidiser particle size threshold: 1

Max inter-pocket kinetic-flame heat flux, W/m²: 1E+09

Max skeleton kinetic-flame heat flux, W/m²: 1E+08

Max out-skeleton kinetic-flame heat flux, W/m²: 1E+08

Parameter bounds (18 base parameters)

ADecompose: [1, 1E+15]

EDecompose: [50000, 300000]

AKineticFlameInterPocket: [100000, 1E+13]

EKineticFlameInterPocket: [50000, 250000]
AKineticFlamePocketOutSkeleton: [100000, 1E+13]
EKineticFlamePocketOutSkeleton: [50000, 250000]
AKineticFlamePocketSkeleton: [100000, 1E+13]
EKineticFlamePocketSkeleton: [50000, 250000]
NuInterPocket: [0, 2.5]
NuPocketOutSkeleton: [0, 2.5]
NuPocketSkeleton: [0, 2.5]
AMetalBurningConstant: [1E-10, 0.001]
BMetalBurningConstant: [1E-12, 1E-05]
DeltaH: [-1E+07, 1E+07]
KDiffusionHeight: [0.001, 10]
APowOrder: [0, 3]
BPowOrder: [0, 3]
KCoefficientRadiationTemperature: [0, 1]

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 100.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 384

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 32

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Values below are display-rounded — not for replay; use the Reproduction Vector block.

Each parameter shows its bounds and a rail flag: [RAILED@lo]/[RAILED@hi] means the gene sits within 0.1% of a bound (fit is bound-limited — consider widening); [interior] means it does not.

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 632668.5770781527

Parameter bounds [1.000E+005; 1.000E+013] [interior]

Activation energy for gas phase reactions in inter-pocket medium 50000.39929523201

Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 4246911.383689705

Parameter bounds [1.000E+005; 1.000E+013] [interior]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.12837437644

Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Reaction order for gas-phase chemistry in inter-pocket medium 2.4999976367684855
Parameter bounds [0.000E+000; 2.500E+000] [RAILED@hi]
Reaction order for gas-phase chemistry outside skeleton structure 0.8219310054776224
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Enthalpy difference between vaporization and binder decomposition -367227.37469848205
Parameter bounds [-1.000E+007; 1.000E+007] [interior]
Diffusion flame height correlation coefficient 0.7654
Parameter bounds [1.000E-003; 1.000E+001] [interior]
Power law exponent for burning rate dependency on A (skeleton thickness) 1.5939
Parameter bounds [0.000E+000; 3.000E+000] [interior]
Power law exponent for burning rate dependency on B (pore diameter) 0.3527
Parameter bounds [0.000E+000; 3.000E+000] [interior]
Radiation temperature correlation coefficient (average layer temperature) 0.0007
Parameter bounds [0.000E+000; 1.000E+000] [RAILED@lo]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 42976750.11973485
Parameter bounds [1.000E+000; 1.000E+015] [interior]
Activation energy for condensed phase reactions 78157.66305319587
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 783736.9153777388
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 50000.0016585601
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Reaction order for gas-phase chemistry in skeleton structure 2.1519130591792806
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 3.366006758096467E-09
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.4665642614130162E-11
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999991775636980.5
Parameter bounds [1.000E+000; 1.000E+015] [RAILED@hi]
Activation energy for condensed phase reactions 164526.31926282833
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 2235154804417.352
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 182531.5148027509
Parameter bounds [5.000E+004; 2.500E+005] [interior]
Reaction order for gas-phase chemistry in skeleton structure 2.015364507350454
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 2.9089004054711024E-08
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0050771647196075E-06
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999975655458555.6
Parameter bounds [1.000E+000; 1.000E+015] [RAILED@hi]
Activation energy for condensed phase reactions 223910.31858146982
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 18868639.84366094
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 78629.56062716022
Parameter bounds [5.000E+004; 2.500E+005] [interior]
Reaction order for gas-phase chemistry in skeleton structure 0.3016702544373872
Parameter bounds [0.000E+000; 2.500E+000] [interior]

Coefficient A for metal combustion heat flux density in skeleton structure, 2.0776103403418784E-09
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 2.889605319801953E-07
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Reproduction Vector

Full-precision result vector. Copy the block below (including the '#' lines) into a text file and replay this run exactly with: --forward-eval <file>

```
# reproduction vector (32 params)
# objective (aggregated fitness) = 0.022782609919312777
# penalty (total aggregated) = 0
# input file = propellants.01234.json
# population = 384
# generated = 2026-08-03 09:41
# layout = shared[0-10], Bas_2[11-17], Bas_1[18-24], Bas_0[25-31]
# count = 32
# checksum = d70c0d14087579312240baaf9ffc17d4e95211dfd7bb7152687978a6162bd3c9
632668.5770781527
50000.39929523201
4246911.383689705
50000.12837437644
2.4999976367684855
0.8219310054776224
-367227.37469848205
0.7654033534384912
1.5939109827048443
0.35272202134926967
0.0007223123187327918
42976750.11973485
78157.66305319587
783736.9153777388
50000.0016585601
2.1519130591792806
3.366006758096467E-09
1.4665642614130162E-11
999991775636980.5
164526.31926282833
2235154804417.352
182531.5148027509
2.015364507350454
2.9089004054711024E-08
1.0050771647196075E-06
999975655458555.6
223910.31858146982
18868639.84366094
78629.56062716022
0.3016702544373872
2.0776103403418784E-09
2.889605319801953E-07
```

Optimization Results

Objective function value 0.0228
Total penalty from constraint violations 0.0000

Burn-Rate Error Breakdown

*Per-point relative error $e = (\text{calc/exp} - 1) * 100\%$. A fuel's RMS is its contribution to the objective (sqrt of the mean squared fractional error over all pressures); a composition objective is the mean of its per-fuel RMS,*

and the overall objective is the mean of the three compositions. Numbers below reflect this report's input vector.

Bas_2 (includes Bas_3, Bas_4) - Burn-Rate Errors

Bas_2

p = 1 MPa: exp 16.9441, calc 13.9367 mm/s -> err -17.75%
p = 1.611 MPa: exp 21.7132, calc 18.5746 mm/s -> err -14.45%
p = 2.222 MPa: exp 25.6654, calc 22.7146 mm/s -> err -11.50%
p = 2.833 MPa: exp 29.1215, calc 26.5297 mm/s -> err -8.90%
p = 3.444 MPa: exp 32.2345, calc 30.1142 mm/s -> err -6.58%
p = 4.056 MPa: exp 35.0917, calc 33.5251 mm/s -> err -4.46%
p = 4.667 MPa: exp 37.7487, calc 36.8000 mm/s -> err -2.51%
p = 5.278 MPa: exp 40.2432, calc 39.9650 mm/s -> err -0.69%
p = 5.889 MPa: exp 42.6026, calc 43.0388 mm/s -> err +1.02%
p = 6.5 MPa: exp 44.8470, calc 46.0355 mm/s -> err +2.65%

Per-fuel RMS = 0.0902 (9.02%)

Bas_3

p = 1 MPa: exp 15.2411, calc 15.4014 mm/s -> err +1.05%
p = 1.611 MPa: exp 20.6812, calc 20.3823 mm/s -> err -1.45%
p = 2.222 MPa: exp 25.4073, calc 24.8787 mm/s -> err -2.08%
p = 2.833 MPa: exp 29.6815, calc 29.1852 mm/s -> err -1.67%
p = 3.444 MPa: exp 33.6334, calc 33.3891 mm/s -> err -0.73%
p = 4.056 MPa: exp 37.3394, calc 37.5189 mm/s -> err +0.48%
p = 4.667 MPa: exp 40.8488, calc 41.5842 mm/s -> err +1.80%
p = 5.278 MPa: exp 44.1961, calc 45.5800 mm/s -> err +3.13%
p = 5.889 MPa: exp 47.4063, calc 49.5127 mm/s -> err +4.44%
p = 6.5 MPa: exp 50.4986, calc 53.3834 mm/s -> err +5.71%

Per-fuel RMS = 0.0277 (2.77%)

Bas_4

p = 1 MPa: exp 9.7279, calc 11.5106 mm/s -> err +18.33%
p = 1.611 MPa: exp 13.5834, calc 14.1569 mm/s -> err +4.22%
p = 2.222 MPa: exp 17.0126, calc 16.8110 mm/s -> err -1.19%
p = 2.833 MPa: exp 20.1664, calc 19.5155 mm/s -> err -3.23%
p = 3.444 MPa: exp 23.1208, calc 22.2522 mm/s -> err -3.76%
p = 4.056 MPa: exp 25.9212, calc 25.0047 mm/s -> err -3.54%
p = 4.667 MPa: exp 28.5972, calc 27.7626 mm/s -> err -2.92%
p = 5.278 MPa: exp 31.1699, calc 30.5195 mm/s -> err -2.09%
p = 5.889 MPa: exp 33.6545, calc 33.2723 mm/s -> err -1.14%
p = 6.5 MPa: exp 36.0627, calc 36.0189 mm/s -> err -0.12%

Per-fuel RMS = 0.0637 (6.37%)

Composition objective (mean per-fuel RMS) = 0.0605 (6.05%)

Bas_1 - Burn-Rate Errors

Bas_1

p = 1 MPa: exp 14.2647, calc 14.4593 mm/s -> err +1.36%
p = 1.611 MPa: exp 19.9183, calc 19.9228 mm/s -> err +0.02%
p = 2.222 MPa: exp 24.9468, calc 24.8568 mm/s -> err -0.36%
p = 2.833 MPa: exp 29.5714, calc 29.4324 mm/s -> err -0.47%
p = 3.444 MPa: exp 33.9037, calc 33.7481 mm/s -> err -0.46%
p = 4.056 MPa: exp 38.0101, calc 37.8657 mm/s -> err -0.38%
p = 4.667 MPa: exp 41.9342, calc 41.8267 mm/s -> err -0.26%
p = 5.278 MPa: exp 45.7067, calc 45.6604 mm/s -> err -0.10%
p = 5.889 MPa: exp 49.3500, calc 49.3882 mm/s -> err +0.08%
p = 6.5 MPa: exp 52.8815, calc 53.0263 mm/s -> err +0.27%

Per-fuel RMS = 0.0052 (0.52%)

Composition objective (mean per-fuel RMS) = 0.0052 (0.52%)

Bas_0 - Burn-Rate Errors

Bas_0

p = 1 MPa: exp 5.9659, calc 5.9831 mm/s -> err +0.29%
 p = 1.611 MPa: exp 7.9046, calc 7.8646 mm/s -> err -0.51%
 p = 2.222 MPa: exp 9.5561, calc 9.5301 mm/s -> err -0.27%
 p = 2.833 MPa: exp 11.0289, calc 11.0320 mm/s -> err +0.03%
 p = 3.444 MPa: exp 12.3759, calc 12.4025 mm/s -> err +0.22%
 p = 4.056 MPa: exp 13.6278, calc 13.6662 mm/s -> err +0.28%
 p = 4.667 MPa: exp 14.8044, calc 14.8413 mm/s -> err +0.25%
 p = 5.278 MPa: exp 15.9192, calc 15.9418 mm/s -> err +0.14%
 p = 5.889 MPa: exp 16.9822, calc 16.9783 mm/s -> err -0.02%
 p = 6.5 MPa: exp 18.0009, calc 17.9595 mm/s -> err -0.23%

Per-fuel RMS = 0.0026 (0.26%)

Composition objective (mean per-fuel RMS) = 0.0026 (0.26%)

Overall objective (mean of the three compositions) = 0.0228 (2.28%)

Vieille Power-Law Fit ($U = A \cdot p^\nu$, U in m/s, p in Pa)

Experimental A/ν are the supplied Vieille coefficients (the experimental burn rate is defined as exactly $A \cdot p^\nu$, so they are exact, not re-fitted). Calculated A/ν are an ordinary-least-squares fit of the model burn rates in log-log space; R^2 shows how power-law-like the model curve is, and $dv = \nu_{\text{calc}} - \nu_{\text{exp}}$ is the model's error in the pressure exponent. Numbers below reflect this report's input vector.

Bas_2 (includes Bas_3, Bas_4) - Vieille Coefficients

Bas_2

exp: $A = 1.2853\text{E-}005$, $\nu = 0.5200$
 calc: $A = 1.9682\text{E-}006$, $\nu = 0.6406$ ($R^2 = 0.9996$); $dv = +0.1206$

Bas_3

exp: $A = 2.2030\text{E-}006$, $\nu = 0.6400$
 calc: $A = 1.4488\text{E-}006$, $\nu = 0.6686$ ($R^2 = 0.9976$); $dv = +0.0286$

Bas_4

exp: $A = 6.1379\text{E-}007$, $\nu = 0.7000$
 calc: $A = 1.9820\text{E-}006$, $\nu = 0.6220$ ($R^2 = 0.9881$); $dv = -0.0780$

Bas_1 - Vieille Coefficients

Bas_1

exp: $A = 9.0004\text{E-}007$, $\nu = 0.7000$
 calc: $A = 9.6014\text{E-}007$, $\nu = 0.6957$ ($R^2 = 0.9999$); $dv = -0.0043$

Bas_0 - Vieille Coefficients

Bas_0

exp: $A = 1.7206\text{E-}006$, $\nu = 0.5900$
 calc: $A = 1.7071\text{E-}006$, $\nu = 0.5905$ ($R^2 = 0.9999$); $dv = +0.0005$

Flame Structure & Heat-Flux Decomposition

q = total heat feedback to the surface (MW/m²). It splits exactly into three competing-flame pathways whose shares sum to 100%: out-kin = out-of-skeleton kinetic flame; skeleton = surface-weighted (metal burning + skeleton kinetic flame in the pores); diff = diffusion flame. h = flame heights (um). T_{s_p} / T_{s_i} = pocket / inter-pocket surface temperature (the two-temperature surface). Mean shares are the pressure-average per fuel. Numbers below reflect this report's input vector.

Bas_2 (includes Bas_3, Bas_4) - Flame Structure

Bas_2

p 1 MPa: $q = 1.44$ MW/m²; out-kin 49.4%, skeleton 31.1%, diff 19.6%
 $h[\text{um}]$ skel-kin 223.9, out-kin 70.7, diff 1061.5; T_{s_p} 623 K, T_{s_i} 643 K
 p 1.611 MPa: $q = 2.05$ MW/m²; out-kin 50.7%, skeleton 38.5%, diff 10.7%
 $h[\text{um}]$ skel-kin 100.4, out-kin 60.0, diff 1347.8; T_{s_p} 633 K, T_{s_i} 665 K
 p 2.222 MPa: $q = 2.65$ MW/m²; out-kin 49.0%, skeleton 44.0%, diff 7.0%
 $h[\text{um}]$ skel-kin 59.2, out-kin 54.5, diff 1605.7; T_{s_p} 640 K, T_{s_i} 681 K
 p 2.833 MPa: $q = 3.23$ MW/m²; out-kin 46.7%, skeleton 48.3%, diff 5.0%
 $h[\text{um}]$ skel-kin 40.0, out-kin 50.9, diff 1843.8; T_{s_p} 646 K, T_{s_i} 694 K

p 3.444 MPa: q = 3.80 MW/m²; out-kin 44.4%, skeleton 51.9%, diff 3.8%
h[um] skel-kin 29.2, out-kin 48.3, diff 2067.5; Ts_p 652 K, Ts_i 705 K
p 4.056 MPa: q = 4.37 MW/m²; out-kin 42.1%, skeleton 54.9%, diff 3.0%
h[um] skel-kin 22.5, out-kin 46.4, diff 2280.4; Ts_p 656 K, Ts_i 715 K
p 4.667 MPa: q = 4.93 MW/m²; out-kin 40.1%, skeleton 57.5%, diff 2.4%
h[um] skel-kin 18.0, out-kin 44.9, diff 2484.7; Ts_p 660 K, Ts_i 723 K
p 5.278 MPa: q = 5.48 MW/m²; out-kin 38.2%, skeleton 59.8%, diff 2.0%
h[um] skel-kin 14.9, out-kin 43.6, diff 2682.2; Ts_p 664 K, Ts_i 730 K
p 5.889 MPa: q = 6.03 MW/m²; out-kin 36.5%, skeleton 61.9%, diff 1.7%
h[um] skel-kin 12.5, out-kin 42.6, diff 2873.9; Ts_p 667 K, Ts_i 737 K
p 6.5 MPa: q = 6.57 MW/m²; out-kin 34.9%, skeleton 63.7%, diff 1.5%
h[um] skel-kin 10.7, out-kin 41.7, diff 3060.8; Ts_p 670 K, Ts_i 743 K
mean shares: out-kin 43.2%, skeleton 51.2%, diffusion 5.7%

Bas_3

p 1 MPa: q = 2.87 MW/m²; out-kin 40.6%, skeleton 14.0%, diff 45.4%
h[um] skel-kin 143.6, out-kin 110.7, diff 232.8; Ts_p 643 K, Ts_i 643 K
p 1.611 MPa: q = 3.53 MW/m²; out-kin 42.0%, skeleton 26.3%, diff 31.8%
h[um] skel-kin 59.3, out-kin 85.9, diff 269.3; Ts_p 649 K, Ts_i 665 K
p 2.222 MPa: q = 4.26 MW/m²; out-kin 39.8%, skeleton 37.2%, diff 23.0%
h[um] skel-kin 33.8, out-kin 74.9, diff 307.5; Ts_p 655 K, Ts_i 681 K
p 2.833 MPa: q = 5.04 MW/m²; out-kin 36.6%, skeleton 46.2%, diff 17.2%
h[um] skel-kin 22.6, out-kin 68.7, diff 346.7; Ts_p 661 K, Ts_i 694 K
p 3.444 MPa: q = 5.86 MW/m²; out-kin 33.4%, skeleton 53.3%, diff 13.3%
h[um] skel-kin 16.5, out-kin 64.9, diff 386.3; Ts_p 666 K, Ts_i 705 K
p 4.056 MPa: q = 6.70 MW/m²; out-kin 30.5%, skeleton 59.0%, diff 10.5%
h[um] skel-kin 12.7, out-kin 62.2, diff 425.8; Ts_p 670 K, Ts_i 715 K
p 4.667 MPa: q = 7.56 MW/m²; out-kin 28.0%, skeleton 63.5%, diff 8.5%
h[um] skel-kin 10.3, out-kin 60.3, diff 465.1; Ts_p 675 K, Ts_i 723 K
p 5.278 MPa: q = 8.44 MW/m²; out-kin 25.8%, skeleton 67.1%, diff 7.0%
h[um] skel-kin 8.5, out-kin 58.8, diff 503.8; Ts_p 679 K, Ts_i 730 K
p 5.889 MPa: q = 9.31 MW/m²; out-kin 24.0%, skeleton 70.1%, diff 5.9%
h[um] skel-kin 7.2, out-kin 57.6, diff 541.9; Ts_p 682 K, Ts_i 737 K
p 6.5 MPa: q = 10.20 MW/m²; out-kin 22.4%, skeleton 72.6%, diff 5.1%
h[um] skel-kin 6.2, out-kin 56.6, diff 579.4; Ts_p 686 K, Ts_i 743 K
mean shares: out-kin 32.3%, skeleton 50.9%, diffusion 16.8%

Bas_4

p 1 MPa: q = 0.58 MW/m²; out-kin 5.7%, skeleton 45.8%, diff 48.4%
h[um] skel-kin 255.0, out-kin 660.2, diff 1063.5; Ts_p 600 K, Ts_i 643 K
p 1.611 MPa: q = 0.73 MW/m²; out-kin 6.4%, skeleton 60.2%, diff 33.3%
h[um] skel-kin 103.8, out-kin 502.5, diff 1223.9; Ts_p 605 K, Ts_i 665 K
p 2.222 MPa: q = 0.92 MW/m²; out-kin 6.1%, skeleton 70.9%, diff 23.0%
h[um] skel-kin 59.1, out-kin 435.2, diff 1411.4; Ts_p 611 K, Ts_i 681 K
p 2.833 MPa: q = 1.13 MW/m²; out-kin 5.5%, skeleton 78.2%, diff 16.3%
h[um] skel-kin 39.5, out-kin 398.8, diff 1612.0; Ts_p 616 K, Ts_i 694 K
p 3.444 MPa: q = 1.36 MW/m²; out-kin 4.9%, skeleton 83.1%, diff 12.0%
h[um] skel-kin 29.0, out-kin 376.0, diff 1819.0; Ts_p 621 K, Ts_i 705 K
p 4.056 MPa: q = 1.60 MW/m²; out-kin 4.3%, skeleton 86.6%, diff 9.1%
h[um] skel-kin 22.5, out-kin 360.4, diff 2029.2; Ts_p 626 K, Ts_i 715 K
p 4.667 MPa: q = 1.86 MW/m²; out-kin 3.9%, skeleton 89.0%, diff 7.1%
h[um] skel-kin 18.2, out-kin 349.0, diff 2240.7; Ts_p 630 K, Ts_i 723 K
p 5.278 MPa: q = 2.12 MW/m²; out-kin 3.5%, skeleton 90.9%, diff 5.7%
h[um] skel-kin 15.2, out-kin 340.3, diff 2452.7; Ts_p 634 K, Ts_i 730 K
p 5.889 MPa: q = 2.40 MW/m²; out-kin 3.1%, skeleton 92.2%, diff 4.6%
h[um] skel-kin 12.9, out-kin 333.4, diff 2664.7; Ts_p 637 K, Ts_i 737 K
p 6.5 MPa: q = 2.68 MW/m²; out-kin 2.9%, skeleton 93.3%, diff 3.8%
h[um] skel-kin 11.2, out-kin 327.7, diff 2876.3; Ts_p 641 K, Ts_i 743 K
mean shares: out-kin 4.6%, skeleton 79.0%, diffusion 16.3%

Bas_1 - Flame Structure

Bas_1

p 1 MPa: q = 1.38 MW/m²; out-kin 50.6%, skeleton 29.1%, diff 20.3%
h[um] skel-kin 179.2, out-kin 72.9, diff 1088.6; Ts_p 618 K, Ts_i 628 K
p 1.611 MPa: q = 1.94 MW/m²; out-kin 51.3%, skeleton 37.7%, diff 11.0%
h[um] skel-kin 86.2, out-kin 64.4, diff 1430.0; Ts_p 623 K, Ts_i 639 K
p 2.222 MPa: q = 2.48 MW/m²; out-kin 49.3%, skeleton 43.6%, diff 7.1%
h[um] skel-kin 53.2, out-kin 59.8, diff 1739.3; Ts_p 627 K, Ts_i 647 K
p 2.833 MPa: q = 2.99 MW/m²; out-kin 46.9%, skeleton 48.1%, diff 5.0%
h[um] skel-kin 37.0, out-kin 56.9, diff 2025.7; Ts_p 630 K, Ts_i 652 K
p 3.444 MPa: q = 3.48 MW/m²; out-kin 44.5%, skeleton 51.7%, diff 3.8%
h[um] skel-kin 27.8, out-kin 54.8, diff 2295.4; Ts_p 633 K, Ts_i 657 K
p 4.056 MPa: q = 3.97 MW/m²; out-kin 42.3%, skeleton 54.7%, diff 3.0%
h[um] skel-kin 21.8, out-kin 53.1, diff 2552.3; Ts_p 635 K, Ts_i 661 K
p 4.667 MPa: q = 4.44 MW/m²; out-kin 40.2%, skeleton 57.3%, diff 2.4%
h[um] skel-kin 17.8, out-kin 51.8, diff 2799.2; Ts_p 637 K, Ts_i 665 K
p 5.278 MPa: q = 4.90 MW/m²; out-kin 38.4%, skeleton 59.6%, diff 2.0%
h[um] skel-kin 14.9, out-kin 50.7, diff 3037.8; Ts_p 638 K, Ts_i 668 K
p 5.889 MPa: q = 5.36 MW/m²; out-kin 36.7%, skeleton 61.6%, diff 1.7%
h[um] skel-kin 12.7, out-kin 49.8, diff 3269.7; Ts_p 640 K, Ts_i 671 K
p 6.5 MPa: q = 5.81 MW/m²; out-kin 35.2%, skeleton 63.4%, diff 1.5%
h[um] skel-kin 11.0, out-kin 49.0, diff 3495.9; Ts_p 641 K, Ts_i 673 K
mean shares: out-kin 43.5%, skeleton 50.7%, diffusion 5.8%

Bas_0 - Flame Structure

Bas_0

p 1 MPa: q = 1.98 MW/m²; out-kin 56.7%, skeleton 9.8%, diff 33.5%
h[um] skel-kin 460.2, out-kin 23.5, diff 428.0; Ts_p 817 K, Ts_i 837 K
p 1.611 MPa: q = 2.54 MW/m²; out-kin 71.5%, skeleton 7.6%, diff 20.9%
h[um] skel-kin 490.7, out-kin 19.8, diff 536.4; Ts_p 823 K, Ts_i 853 K
p 2.222 MPa: q = 3.05 MW/m²; out-kin 79.0%, skeleton 6.4%, diff 14.7%
h[um] skel-kin 520.3, out-kin 17.9, diff 635.0; Ts_p 827 K, Ts_i 863 K
p 2.833 MPa: q = 3.52 MW/m²; out-kin 83.3%, skeleton 5.6%, diff 11.1%
h[um] skel-kin 545.9, out-kin 16.7, diff 724.4; Ts_p 830 K, Ts_i 871 K
p 3.444 MPa: q = 3.96 MW/m²; out-kin 86.0%, skeleton 5.1%, diff 8.9%
h[um] skel-kin 567.9, out-kin 15.8, diff 806.2; Ts_p 833 K, Ts_i 878 K
p 4.056 MPa: q = 4.36 MW/m²; out-kin 87.9%, skeleton 4.8%, diff 7.4%
h[um] skel-kin 587.0, out-kin 15.1, diff 881.6; Ts_p 835 K, Ts_i 883 K
p 4.667 MPa: q = 4.74 MW/m²; out-kin 89.2%, skeleton 4.5%, diff 6.3%
h[um] skel-kin 603.6, out-kin 14.5, diff 951.6; Ts_p 837 K, Ts_i 888 K
p 5.278 MPa: q = 5.10 MW/m²; out-kin 90.2%, skeleton 4.3%, diff 5.5%
h[um] skel-kin 618.4, out-kin 13.9, diff 1017.2; Ts_p 839 K, Ts_i 893 K
p 5.889 MPa: q = 5.43 MW/m²; out-kin 91.0%, skeleton 4.2%, diff 4.8%
h[um] skel-kin 631.6, out-kin 13.5, diff 1078.9; Ts_p 841 K, Ts_i 896 K
p 6.5 MPa: q = 5.75 MW/m²; out-kin 91.7%, skeleton 4.0%, diff 4.3%
h[um] skel-kin 643.5, out-kin 13.1, diff 1137.4; Ts_p 842 K, Ts_i 900 K
mean shares: out-kin 82.6%, skeleton 5.6%, diffusion 11.7%

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s
Calculated burning rate 13.94 mm/s

First Conditional Fuel

Surface temperature 642.77 K
Linear burning rate 11.54 mm/s
Mass decomposition rate 19.14 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.87e+06 W/m²
Kinetic flame height 106.78 μm
Average kinetic flame density 1.82 kg/m³

Average kinetic flame temperature 2,174.89 K
Binder sublimation heat flux density 2.87e+06 W/m²
Surface heat balance error -7.6e-05 W/m²

Secondary Conditional Fuel

Surface temperature 622.79 K
Linear burning rate 7.22 mm/s
Mass decomposition rate 11.97 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.46e+06 W/m²
Kinetic flame height 70.72 μm
Average kinetic flame density 2.66 kg/m³
Average kinetic flame temperature 1,493.9 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 432,409.51 W/m²
Kinetic flame height 223.91 μm
Average kinetic flame density 3.59 kg/m³
Average kinetic flame temperature 1,106.9 K
Heat flux density due to metal combustion, 193,963.72 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 1.34e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 8.72e-06 m
Pore diameter 8.35e-11 m

Diffusion flame heat flux density 280,851.6 W/m²
Diffusion flame height 1,061.49 μm
Heat flux density to surface within skeletal structure 446,005.48 W/m²
Heat flux density to surface outside skeletal structure 709,383.65 W/m²
Total heat flux density to surface 1.44e+06 W/m²
Binder sublimation heat flux density 1.44e+06 W/m²
Surface heat balance error 1.46e-04 W/m²

At pressure 1.61 MPa

Experimental burning rate 21.71 mm/s
Calculated burning rate 18.57 mm/s

First Conditional Fuel

Surface temperature 665.16 K
Linear burning rate 18.87 mm/s
Mass decomposition rate 31.31 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 5.75e+06 W/m²
Kinetic flame height 52.95 μm
Average kinetic flame density 2.93 kg/m³
Average kinetic flame temperature 2,186.08 K
Binder sublimation heat flux density 5.75e+06 W/m²
Surface heat balance error 5.66e-04 W/m²

Secondary Conditional Fuel

Surface temperature 632.81 K
Linear burning rate 9.16 mm/s
Mass decomposition rate 15.2 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.89e+06 W/m²
Kinetic flame height 60.03 μm
Average kinetic flame density 4.27 kg/m³
Average kinetic flame temperature 1,498.9 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 954,529.57 W/m²
Kinetic flame height 100.38 μm
Average kinetic flame density 5.75 kg/m³
Average kinetic flame temperature 1,111.9 K

Heat flux density due to metal combustion, 282,117.41 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 1.23e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.96e-06 m
Pore diameter 7.68e-11 m
Diffusion flame heat flux density 220,446.2 W/m²
Diffusion flame height 1,347.81 μm
Heat flux density to surface within skeletal structure 790,514.66 W/m²
Heat flux density to surface outside skeletal structure 1.04e+06 W/m²
Total heat flux density to surface 2.05e+06 W/m²
Binder sublimation heat flux density 2.05e+06 W/m²
Surface heat balance error -3.29e-04 W/m²

At pressure 2.22 MPa

Experimental burning rate 25.67 mm/s
Calculated burning rate 22.71 mm/s

First Conditional Fuel

Surface temperature 681.42 K
Linear burning rate 26.44 mm/s
Mass decomposition rate 43.87 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 9.12e+06 W/m²
Kinetic flame height 33.17 μm
Average kinetic flame density 4.02 kg/m³
Average kinetic flame temperature 2,194.21 K
Binder sublimation heat flux density 9.12e+06 W/m²
Surface heat balance error -0.0015 W/m²

Secondary Conditional Fuel

Surface temperature 640.35 K
Linear burning rate 10.92 mm/s
Mass decomposition rate 18.11 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.17e+06 W/m²
Kinetic flame height 54.47 μm
Average kinetic flame density 5.87 kg/m³
Average kinetic flame temperature 1,502.68 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.61e+06 W/m²
Kinetic flame height 59.21 μm
Average kinetic flame density 7.91 kg/m³
Average kinetic flame temperature 1,115.68 K
Heat flux density due to metal combustion, 371,234.09 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 1.16e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.51e-06 m
Pore diameter 7.22e-11 m
Diffusion flame heat flux density 184,572.1 W/m²
Diffusion flame height 1,605.68 μm
Heat flux density to surface within skeletal structure 1.17e+06 W/m²
Heat flux density to surface outside skeletal structure 1.3e+06 W/m²
Total heat flux density to surface 2.65e+06 W/m²
Binder sublimation heat flux density 2.65e+06 W/m²
Surface heat balance error -5.92e-04 W/m²

At pressure 2.83 MPa

Experimental burning rate 29.12 mm/s

Calculated burning rate 26.53 mm/s

First Conditional Fuel

Surface temperature 694.37 K
Linear burning rate 34.2 mm/s
Mass decomposition rate $56.74 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $1.29 \times 10^7 \text{ W/m}^2$
Kinetic flame height 23.36 μm
Average kinetic flame density 5.11 kg/m^3
Average kinetic flame temperature 2,200.69 K
Binder sublimation heat flux density $1.29 \times 10^7 \text{ W/m}^2$
Surface heat balance error 0.0016 W/m^2

Secondary Conditional Fuel

Surface temperature 646.44 K
Linear burning rate 12.53 mm/s
Mass decomposition rate $20.8 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.38 \times 10^6 \text{ W/m}^2$
Kinetic flame height 50.89 μm
Average kinetic flame density 7.47 kg/m^3
Average kinetic flame temperature 1,505.72 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.36 \times 10^6 \text{ W/m}^2$
Kinetic flame height 39.95 μm
Average kinetic flame density 10.05 kg/m^3
Average kinetic flame temperature 1,118.72 K
Heat flux density due to metal combustion, $461,056.91 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $1.1 \times 10^{-7} \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $3.62 \times 10^{-6} \text{ m}$
Pore diameter $6.87 \times 10^{-11} \text{ m}$

Diffusion flame heat flux density $160,409.55 \text{ W/m}^2$
Diffusion flame height 1,843.75 μm
Heat flux density to surface within skeletal structure $1.56 \times 10^6 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.51 \times 10^6 \text{ W/m}^2$
Total heat flux density to surface $3.23 \times 10^6 \text{ W/m}^2$
Binder sublimation heat flux density $3.23 \times 10^6 \text{ W/m}^2$
Surface heat balance error $4.59 \times 10^{-6} \text{ W/m}^2$

At pressure 3.44 MPa

Experimental burning rate 32.23 mm/s

Calculated burning rate 30.11 mm/s

First Conditional Fuel

Surface temperature 705.21 K
Linear burning rate 42.11 mm/s
Mass decomposition rate $69.86 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $1.7 \times 10^7 \text{ W/m}^2$
Kinetic flame height 17.64 μm
Average kinetic flame density 6.2 kg/m^3
Average kinetic flame temperature 2,206.11 K
Binder sublimation heat flux density $1.7 \times 10^7 \text{ W/m}^2$
Surface heat balance error $2.25 \times 10^{-4} \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 651.57 K
Linear burning rate 14.06 mm/s
Mass decomposition rate $23.32 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.54 \times 10^6 \text{ W/m}^2$

Kinetic flame height 48.34 μm
Average kinetic flame density 9.06 kg/m^3
Average kinetic flame temperature 1,508.29 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.22e+06 W/m^2
Kinetic flame height 29.21 μm
Average kinetic flame density 12.19 kg/m^3
Average kinetic flame temperature 1,121.29 K
Heat flux density due to metal combustion, 551,680.44 W/m^2
Effective thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 1.06e-07 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.01e-06 m
Pore diameter 6.6e-11 m
Diffusion flame heat flux density 142,802.16 W/m^2
Diffusion flame height 2,067.49 μm
Heat flux density to surface within skeletal structure 1.97e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.69e+06 W/m^2
Total heat flux density to surface 3.8e+06 W/m^2
Binder sublimation heat flux density 3.8e+06 W/m^2
Surface heat balance error -1.33e-04 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 33.53 mm/s

First Conditional Fuel

Surface temperature 714.58 K
Linear burning rate 50.15 mm/s
Mass decomposition rate 83.2 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 2.14e+07 W/m^2
Kinetic flame height 13.96 μm
Average kinetic flame density 7.28 kg/m^3
Average kinetic flame temperature 2,210.79 K
Binder sublimation heat flux density 2.14e+07 W/m^2
Surface heat balance error -7.84e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 656.03 K
Linear burning rate 15.5 mm/s
Mass decomposition rate 25.72 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.68e+06 W/m^2
Kinetic flame height 46.41 μm
Average kinetic flame density 10.66 kg/m^3
Average kinetic flame temperature 1,510.52 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.15e+06 W/m^2
Kinetic flame height 22.53 μm
Average kinetic flame density 14.33 kg/m^3
Average kinetic flame temperature 1,123.52 K
Heat flux density due to metal combustion, 643,212.93 W/m^2
Effective thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 1.02e-07 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.58e-06 m
Pore diameter 6.38e-11 m
Diffusion flame heat flux density 129,275.11 W/m^2
Diffusion flame height 2,280.38 μm

Heat flux density to surface within skeletal structure $2.4\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.84\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $4.37\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $4.37\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-2.25\text{e}-04 \text{ W/m}^2$

At pressure 4.67 MPa

Experimental burning rate 37.75 mm/s
Calculated burning rate 36.8 mm/s

First Conditional Fuel

Surface temperature 722.86 K
Linear burning rate 58.31 mm/s
Mass decomposition rate $96.74 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.61\text{e}+07 \text{ W/m}^2$
Kinetic flame height 11.42 μm
Average kinetic flame density 8.36 kg/m^3
Average kinetic flame temperature 2,214.93 K
Binder sublimation heat flux density $2.61\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0044 W/m^2

Secondary Conditional Fuel

Surface temperature 659.98 K
Linear burning rate 16.89 mm/s
Mass decomposition rate $28.02 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.8\text{e}+06 \text{ W/m}^2$
Kinetic flame height 44.87 μm
Average kinetic flame density 12.25 kg/m^3
Average kinetic flame temperature 1,512.49 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.16\text{e}+06 \text{ W/m}^2$
Kinetic flame height 18.05 μm
Average kinetic flame density 16.46 kg/m^3
Average kinetic flame temperature 1,125.49 K
Heat flux density due to metal combustion, 735,733.73 W/m^2
Effective thermal conductivity $0.4 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity $0.4 \text{ W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity $9.94\text{e}-08 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $2.25\text{e}-06 \text{ m}$
Pore diameter $6.19\text{e}-11 \text{ m}$
Diffusion flame heat flux density 118,483.57 W/m^2
Diffusion flame height 2,484.75 μm
Heat flux density to surface within skeletal structure $2.83\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.97\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $4.93\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $4.93\text{e}+06 \text{ W/m}^2$
Surface heat balance error $3.32\text{e}-04 \text{ W/m}^2$

At pressure 5.28 MPa

Experimental burning rate 40.24 mm/s
Calculated burning rate 39.96 mm/s

First Conditional Fuel

Surface temperature 730.3 K
Linear burning rate 66.57 mm/s
Mass decomposition rate $110.45 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $3.11\text{e}+07 \text{ W/m}^2$
Kinetic flame height 9.58 μm
Average kinetic flame density 9.44 kg/m^3
Average kinetic flame temperature 2,218.65 K

Binder sublimation heat flux density $3.11\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0019 W/m^2

Secondary Conditional Fuel

Surface temperature 663.55 K
Linear burning rate 18.23 mm/s
Mass decomposition rate $30.25 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.9\text{e}+06 \text{ W/m}^2$
Kinetic flame height $43.61 \text{ }\mu\text{m}$
Average kinetic flame density 13.83 kg/m^3
Average kinetic flame temperature $1,514.27 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.24\text{e}+06 \text{ W/m}^2$
Kinetic flame height $14.87 \text{ }\mu\text{m}$
Average kinetic flame density 18.58 kg/m^3
Average kinetic flame temperature $1,127.27 \text{ K}$
Heat flux density due to metal combustion, $829,295.24 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $9.68\text{e}-08 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.99\text{e}-06 \text{ m}$
Pore diameter $6.02\text{e}-11 \text{ m}$
Diffusion flame heat flux density $109,628.34 \text{ W/m}^2$
Diffusion flame height $2,682.2 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $3.28\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $2.09\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $5.48\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $5.48\text{e}+06 \text{ W/m}^2$
Surface heat balance error $8.21\text{e}-04 \text{ W/m}^2$

At pressure 5.89 MPa

Experimental burning rate 42.6 mm/s
Calculated burning rate 43.04 mm/s

First Conditional Fuel

Surface temperature 737.07 K
Linear burning rate 74.93 mm/s
Mass decomposition rate $124.31 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $3.62\text{e}+07 \text{ W/m}^2$
Kinetic flame height $8.2 \text{ }\mu\text{m}$
Average kinetic flame density 10.52 kg/m^3
Average kinetic flame temperature $2,222.04 \text{ K}$
Binder sublimation heat flux density $3.62\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0054 W/m^2

Secondary Conditional Fuel

Surface temperature 666.8 K
Linear burning rate 19.54 mm/s
Mass decomposition rate $32.41 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.99\text{e}+06 \text{ W/m}^2$
Kinetic flame height $42.56 \text{ }\mu\text{m}$
Average kinetic flame density 15.42 kg/m^3
Average kinetic flame temperature $1,515.9 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $7.38\text{e}+06 \text{ W/m}^2$
Kinetic flame height $12.53 \text{ }\mu\text{m}$
Average kinetic flame density 20.7 kg/m^3
Average kinetic flame temperature $1,128.9 \text{ K}$
Heat flux density due to metal combustion, $923,929.4 \text{ W/m}^2$

Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 9.45e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.78e-06 m
Pore diameter 5.88e-11 m
Diffusion flame heat flux density 102,201.66 W/m²
Diffusion flame height 2,873.93 μm
Heat flux density to surface within skeletal structure 3.73e+06 W/m²
Heat flux density to surface outside skeletal structure 2.2e+06 W/m²
Total heat flux density to surface 6.03e+06 W/m²
Binder sublimation heat flux density 6.03e+06 W/m²
Surface heat balance error 7.77e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 46.04 mm/s

First Conditional Fuel

Surface temperature 743.3 K
Linear burning rate 83.37 mm/s
Mass decomposition rate 138.31 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 4.16e+07 W/m²
Kinetic flame height 7.13 μm
Average kinetic flame density 11.59 kg/m³
Average kinetic flame temperature 2,225.15 K
Binder sublimation heat flux density 4.16e+07 W/m²
Surface heat balance error -0.0025 W/m²

Secondary Conditional Fuel

Surface temperature 669.79 K
Linear burning rate 20.81 mm/s
Mass decomposition rate 34.52 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.07e+06 W/m²
Kinetic flame height 41.67 μm
Average kinetic flame density 17 kg/m³
Average kinetic flame temperature 1,517.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 8.57e+06 W/m²
Kinetic flame height 10.75 μm
Average kinetic flame density 22.82 kg/m³
Average kinetic flame temperature 1,130.39 K
Heat flux density due to metal combustion, 1.02e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 9.24e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.61e-06 m
Pore diameter 5.75e-11 m
Diffusion flame heat flux density 95,863.91 W/m²
Diffusion flame height 3,060.81 μm
Heat flux density to surface within skeletal structure 4.19e+06 W/m²
Heat flux density to surface outside skeletal structure 2.29e+06 W/m²
Total heat flux density to surface 6.57e+06 W/m²
Binder sublimation heat flux density 6.57e+06 W/m²
Surface heat balance error 5.49e-04 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 15.4 mm/s

First Conditional Fuel

Surface temperature 642.77 K
Linear burning rate 11.54 mm/s
Mass decomposition rate $19.14 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.87\text{e}+06 \text{ W/m}^2$
Kinetic flame height 106.78 μm
Average kinetic flame density 1.82 kg/m^3
Average kinetic flame temperature 2,174.89 K
Binder sublimation heat flux density $2.87\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-7.6\text{e}-05 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 642.77 K
Linear burning rate 11.54 mm/s
Mass decomposition rate $19.14 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.56\text{e}+06 \text{ W/m}^2$
Kinetic flame height 110.68 μm
Average kinetic flame density 2.64 kg/m^3
Average kinetic flame temperature 1,503.89 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.38\text{e}+06 \text{ W/m}^2$
Kinetic flame height 143.6 μm
Average kinetic flame density 2.43 kg/m^3
Average kinetic flame temperature 1,632.89 K
Heat flux density due to metal combustion, 222,442.1 W/m^2
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $7\text{e}-07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness $4.13\text{e}-06 \text{ m}$
Pore diameter $7.08\text{e}-11 \text{ m}$

Diffusion flame heat flux density $1.3\text{e}+06 \text{ W/m}^2$
Diffusion flame height 232.78 μm
Heat flux density to surface within skeletal structure 402,426.35 W/m^2
Heat flux density to surface outside skeletal structure $1.17\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $2.87\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $2.87\text{e}+06 \text{ W/m}^2$
Surface heat balance error $4.96\text{e}-04 \text{ W/m}^2$

At pressure 1.61 MPa

Experimental burning rate 20.68 mm/s
Calculated burning rate 20.38 mm/s

First Conditional Fuel

Surface temperature 665.16 K
Linear burning rate 18.87 mm/s
Mass decomposition rate $31.31 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $5.75\text{e}+06 \text{ W/m}^2$
Kinetic flame height 52.95 μm
Average kinetic flame density 2.93 kg/m^3
Average kinetic flame temperature 2,186.08 K
Binder sublimation heat flux density $5.75\text{e}+06 \text{ W/m}^2$
Surface heat balance error $5.66\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 649.24 K
Linear burning rate 13.35 mm/s
Mass decomposition rate $22.14 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2\text{e}+06 \text{ W/m}^2$
Kinetic flame height $85.92 \text{ }\mu\text{m}$
Average kinetic flame density 4.24 kg/m^3
Average kinetic flame temperature $1,507.12 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $3.33\text{e}+06 \text{ W/m}^2$
Kinetic flame height $59.34 \text{ }\mu\text{m}$
Average kinetic flame density 3.91 kg/m^3
Average kinetic flame temperature $1,636.12 \text{ K}$
Heat flux density due to metal combustion, $279,462.85 \text{ W/m}^2$
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $6.65\text{e}-07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $3.27\text{e}-06 \text{ m}$
Pore diameter $6.72\text{e}-11 \text{ m}$

Diffusion flame heat flux density $1.12\text{e}+06 \text{ W/m}^2$
Diffusion flame height $269.27 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $927,927.53 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.48\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $3.53\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $3.53\text{e}+06 \text{ W/m}^2$
Surface heat balance error $4.65\text{e}-05 \text{ W/m}^2$

At pressure 2.22 MPa

Experimental burning rate 25.41 mm/s
Calculated burning rate 24.88 mm/s

First Conditional Fuel

Surface temperature 681.42 K
Linear burning rate 26.44 mm/s
Mass decomposition rate $43.87 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.12\text{e}+06 \text{ W/m}^2$
Kinetic flame height $33.17 \text{ }\mu\text{m}$
Average kinetic flame density 4.02 kg/m^3
Average kinetic flame temperature $2,194.21 \text{ K}$
Binder sublimation heat flux density $9.12\text{e}+06 \text{ W/m}^2$
Surface heat balance error -0.0015 W/m^2

Secondary Conditional Fuel

Surface temperature 655.25 K
Linear burning rate 15.24 mm/s
Mass decomposition rate $25.29 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.28\text{e}+06 \text{ W/m}^2$
Kinetic flame height $74.86 \text{ }\mu\text{m}$
Average kinetic flame density 5.84 kg/m^3
Average kinetic flame temperature $1,510.13 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.82\text{e}+06 \text{ W/m}^2$
Kinetic flame height $33.83 \text{ }\mu\text{m}$
Average kinetic flame density 5.38 kg/m^3
Average kinetic flame temperature $1,639.13 \text{ K}$
Heat flux density due to metal combustion, $344,111.05 \text{ W/m}^2$
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $6.34\text{e}-07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $2.65\text{e}-06 \text{ m}$
Pore diameter $6.41\text{e}-11 \text{ m}$

Diffusion flame heat flux density $981,626.67 \text{ W/m}^2$

Diffusion flame height 307.53 μm
Heat flux density to surface within skeletal structure $1.59\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.7\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $4.26\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $4.26\text{e}+06 \text{ W/m}^2$
Surface heat balance error $6.96\text{e}-04 \text{ W/m}^2$

At pressure 2.83 MPa

Experimental burning rate 29.68 mm/s
Calculated burning rate 29.19 mm/s

First Conditional Fuel

Surface temperature 694.37 K
Linear burning rate 34.2 mm/s
Mass decomposition rate $56.74 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.29\text{e}+07 \text{ W/m}^2$
Kinetic flame height 23.36 μm
Average kinetic flame density 5.11 kg/m^3
Average kinetic flame temperature 2,200.69 K
Binder sublimation heat flux density $1.29\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0016 W/m^2

Secondary Conditional Fuel

Surface temperature 660.78 K
Linear burning rate 17.18 mm/s
Mass decomposition rate $28.51 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.48\text{e}+06 \text{ W/m}^2$
Kinetic flame height 68.73 μm
Average kinetic flame density 7.43 kg/m^3
Average kinetic flame temperature 1,512.89 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.7\text{e}+06 \text{ W/m}^2$
Kinetic flame height 22.56 μm
Average kinetic flame density 6.85 kg/m^3
Average kinetic flame temperature 1,641.89 K
Heat flux density due to metal combustion, 415,256.85 W/m^2
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $6.08\text{e}-07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness $2.19\text{e}-06 \text{ m}$
Pore diameter $6.15\text{e}-11 \text{ m}$
Diffusion flame heat flux density 869,013.86 W/m^2
Diffusion flame height 346.74 μm
Heat flux density to surface within skeletal structure $2.33\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.85\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $5.04\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $5.04\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-2.96\text{e}-04 \text{ W/m}^2$

At pressure 3.44 MPa

Experimental burning rate 33.63 mm/s
Calculated burning rate 33.39 mm/s

First Conditional Fuel

Surface temperature 705.21 K
Linear burning rate 42.11 mm/s
Mass decomposition rate $69.86 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.7\text{e}+07 \text{ W/m}^2$
Kinetic flame height 17.64 μm
Average kinetic flame density 6.2 kg/m^3

Average kinetic flame temperature 2,206.11 K
Binder sublimation heat flux density $1.7\text{e}+07 \text{ W/m}^2$
Surface heat balance error $2.25\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 665.84 K
Linear burning rate 19.15 mm/s
Mass decomposition rate $31.76 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.62\text{e}+06 \text{ W/m}^2$
Kinetic flame height 64.88 μm
Average kinetic flame density 9.02 kg/m^3
Average kinetic flame temperature 1,515.42 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.19\text{e}+07 \text{ W/m}^2$
Kinetic flame height 16.47 μm
Average kinetic flame density 8.31 kg/m^3
Average kinetic flame temperature 1,644.42 K
Heat flux density due to metal combustion, 491,785.57 W/m^2
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $5.85\text{e}-07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness $1.84\text{e}-06 \text{ m}$
Pore diameter $5.92\text{e}-11 \text{ m}$
Diffusion flame heat flux density 778,695.88 W/m^2
Diffusion flame height 386.31 μm
Heat flux density to surface within skeletal structure $3.12\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.96\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $5.86\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $5.86\text{e}+06 \text{ W/m}^2$
Surface heat balance error $2.71\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 37.52 mm/s

First Conditional Fuel

Surface temperature 714.58 K
Linear burning rate 50.15 mm/s
Mass decomposition rate $83.2 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.14\text{e}+07 \text{ W/m}^2$
Kinetic flame height 13.96 μm
Average kinetic flame density 7.28 kg/m^3
Average kinetic flame temperature 2,210.79 K
Binder sublimation heat flux density $2.14\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-7.84\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 670.46 K
Linear burning rate 21.1 mm/s
Mass decomposition rate $35.01 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.72\text{e}+06 \text{ W/m}^2$
Kinetic flame height 62.23 μm
Average kinetic flame density 10.61 kg/m^3
Average kinetic flame temperature 1,517.73 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.53\text{e}+07 \text{ W/m}^2$
Kinetic flame height 12.75 μm
Average kinetic flame density 9.77 kg/m^3
Average kinetic flame temperature 1,646.73 K

Heat flux density due to metal combustion, 572,770.7 W/m²
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 5.66e-07 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.58e-06 m
Pore diameter 5.72e-11 m
Diffusion flame heat flux density 705,327.33 W/m²
Diffusion flame height 425.84 µm
Heat flux density to surface within skeletal structure 3.95e+06 W/m²
Heat flux density to surface outside skeletal structure 2.05e+06 W/m²
Total heat flux density to surface 6.7e+06 W/m²
Binder sublimation heat flux density 6.7e+06 W/m²
Surface heat balance error 3.35e-04 W/m²

At pressure 4.67 MPa

Experimental burning rate 40.85 mm/s
Calculated burning rate 41.58 mm/s

First Conditional Fuel

Surface temperature 722.86 K
Linear burning rate 58.31 mm/s
Mass decomposition rate 96.74 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.61e+07 W/m²
Kinetic flame height 11.42 µm
Average kinetic flame density 8.36 kg/m³
Average kinetic flame temperature 2,214.93 K
Binder sublimation heat flux density 2.61e+07 W/m²
Surface heat balance error -0.0044 W/m²

Secondary Conditional Fuel

Surface temperature 674.71 K
Linear burning rate 23.05 mm/s
Mass decomposition rate 38.24 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.8e+06 W/m²
Kinetic flame height 60.3 µm
Average kinetic flame density 12.19 kg/m³
Average kinetic flame temperature 1,519.85 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.9e+07 W/m²
Kinetic flame height 10.27 µm
Average kinetic flame density 11.23 kg/m³
Average kinetic flame temperature 1,648.85 K
Heat flux density due to metal combustion, 657,461.34 W/m²
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 5.48e-07 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.37e-06 m
Pore diameter 5.54e-11 m
Diffusion flame heat flux density 644,900.53 W/m²
Diffusion flame height 465.08 µm
Heat flux density to surface within skeletal structure 4.8e+06 W/m²
Heat flux density to surface outside skeletal structure 2.12e+06 W/m²
Total heat flux density to surface 7.56e+06 W/m²
Binder sublimation heat flux density 7.56e+06 W/m²
Surface heat balance error 0.0016 W/m²

At pressure 5.28 MPa

Experimental burning rate 44.2 mm/s

Calculated burning rate 45.58 mm/s

First Conditional Fuel

Surface temperature 730.3 K
Linear burning rate 66.57 mm/s
Mass decomposition rate $110.45 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $3.11 \text{e}+07 \text{ W/m}^2$
Kinetic flame height 9.58 μm
Average kinetic flame density 9.44 kg/m^3
Average kinetic flame temperature 2,218.65 K
Binder sublimation heat flux density $3.11 \text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0019 W/m^2

Secondary Conditional Fuel

Surface temperature 678.6 K
Linear burning rate 24.97 mm/s
Mass decomposition rate $41.42 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.87 \text{e}+06 \text{ W/m}^2$
Kinetic flame height 58.79 μm
Average kinetic flame density 13.76 kg/m^3
Average kinetic flame temperature 1,521.8 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.28 \text{e}+07 \text{ W/m}^2$
Kinetic flame height 8.52 μm
Average kinetic flame density 12.69 kg/m^3
Average kinetic flame temperature 1,650.8 K
Heat flux density due to metal combustion, 744,980.53 W/m^2
Effective thermal conductivity $0.22 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $5.33 \text{e}-07 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306 \text{E}-08$
Skeleton layer thickness $1.21 \text{e}-06 \text{ m}$
Pore diameter $5.39 \text{e}-11 \text{ m}$

Diffusion flame heat flux density 594,599.04 W/m^2
Diffusion flame height 503.77 μm
Heat flux density to surface within skeletal structure $5.66 \text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $2.18 \text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $8.44 \text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $8.44 \text{e}+06 \text{ W/m}^2$
Surface heat balance error 0.0012 W/m^2

At pressure 5.89 MPa

Experimental burning rate 47.41 mm/s

Calculated burning rate 49.51 mm/s

First Conditional Fuel

Surface temperature 737.07 K
Linear burning rate 74.93 mm/s
Mass decomposition rate $124.31 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $3.62 \text{e}+07 \text{ W/m}^2$
Kinetic flame height 8.2 μm
Average kinetic flame density 10.52 kg/m^3
Average kinetic flame temperature 2,222.04 K
Binder sublimation heat flux density $3.62 \text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0054 W/m^2

Secondary Conditional Fuel

Surface temperature 682.19 K
Linear burning rate 26.86 mm/s
Mass decomposition rate $44.56 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.92 \text{e}+06 \text{ W/m}^2$

Kinetic flame height 57.58 μm
Average kinetic flame density 15.34 kg/m^3
Average kinetic flame temperature 1,523.6 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.68e+07 W/m^2
Kinetic flame height 7.23 μm
Average kinetic flame density 14.14 kg/m^3
Average kinetic flame temperature 1,652.6 K
Heat flux density due to metal combustion, 834,983.81 W/m^2
Effective thermal conductivity 0.22 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.22 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 5.19e-07 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.07e-06 m
Pore diameter 5.25e-11 m

Diffusion flame heat flux density 552,105.62 W/m^2
Diffusion flame height 541.89 μm
Heat flux density to surface within skeletal structure 6.53e+06 W/m^2
Heat flux density to surface outside skeletal structure 2.23e+06 W/m^2
Total heat flux density to surface 9.31e+06 W/m^2
Binder sublimation heat flux density 9.31e+06 W/m^2
Surface heat balance error 0.0019 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 53.38 mm/s

First Conditional Fuel

Surface temperature 743.3 K
Linear burning rate 83.37 mm/s
Mass decomposition rate 138.31 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 4.16e+07 W/m^2
Kinetic flame height 7.13 μm
Average kinetic flame density 11.59 kg/m^3
Average kinetic flame temperature 2,225.15 K
Binder sublimation heat flux density 4.16e+07 W/m^2
Surface heat balance error -0.0025 W/m^2

Secondary Conditional Fuel

Surface temperature 685.52 K
Linear burning rate 28.72 mm/s
Mass decomposition rate 47.64 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.97e+06 W/m^2
Kinetic flame height 56.58 μm
Average kinetic flame density 16.91 kg/m^3
Average kinetic flame temperature 1,525.26 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.1e+07 W/m^2
Kinetic flame height 6.24 μm
Average kinetic flame density 15.6 kg/m^3
Average kinetic flame temperature 1,654.26 K
Heat flux density due to metal combustion, 927,069.23 W/m^2
Effective thermal conductivity 0.22 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.22 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 5.07e-07 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 9.65e-07 m
Pore diameter 5.13e-11 m

Diffusion flame heat flux density 515,789.4 W/m^2
Diffusion flame height 579.4 μm

Heat flux density to surface within skeletal structure $7.4\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $2.28\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.02\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.02\text{e}+07 \text{ W/m}^2$
Surface heat balance error $1.07\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 11.51 mm/s

First Conditional Fuel

Surface temperature 642.77 K
Linear burning rate 11.54 mm/s
Mass decomposition rate $19.14 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.87\text{e}+06 \text{ W/m}^2$
Kinetic flame height 106.78 μm
Average kinetic flame density 1.82 kg/m^3
Average kinetic flame temperature 2,174.89 K
Binder sublimation heat flux density $2.87\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-7.6\text{e}-05 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 600 K
Linear burning rate 4.07 mm/s
Mass decomposition rate $6.75 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 66,946.74 W/m^2
Kinetic flame height 660.23 μm
Average kinetic flame density 4.83 kg/m^3
Average kinetic flame temperature 821 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 251,404.51 W/m^2
Kinetic flame height 254.97 μm
Average kinetic flame density 4.31 kg/m^3
Average kinetic flame temperature 920.5 K
Heat flux density due to metal combustion, 275,437.08 W/m^2
Effective thermal conductivity 1.41 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 1.41 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity $6.23\text{e}-08 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness $2.18\text{e}-05 \text{ m}$
Pore diameter $1.02\text{e}-10 \text{ m}$
Diffusion flame heat flux density 280,194.16 W/m^2
Diffusion flame height 1,063.55 μm
Heat flux density to surface within skeletal structure 265,318.16 W/m^2
Heat flux density to surface outside skeletal structure 33,232.27 W/m^2
Total heat flux density to surface 578,744.59 W/m^2
Binder sublimation heat flux density 578,744.59 W/m^2
Surface heat balance error $9.11\text{e}-05 \text{ W/m}^2$

At pressure 1.61 MPa

Experimental burning rate 13.58 mm/s
Calculated burning rate 14.16 mm/s

First Conditional Fuel

Surface temperature 665.16 K
Linear burning rate 18.87 mm/s
Mass decomposition rate $31.31 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $5.75\text{e}+06 \text{ W/m}^2$
Kinetic flame height 52.95 μm

Average kinetic flame density 2.93 kg/m³
Average kinetic flame temperature 2,186.08 K
Binder sublimation heat flux density 5.75e+06 W/m²
Surface heat balance error 5.66e-04 W/m²

Secondary Conditional Fuel

Surface temperature 605.43 K
Linear burning rate 4.68 mm/s
Mass decomposition rate 7.76 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 86,877.76 W/m²
Kinetic flame height 502.51 μm
Average kinetic flame density 7.76 kg/m³
Average kinetic flame temperature 823.71 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 612,328.97 W/m²
Kinetic flame height 103.8 μm
Average kinetic flame density 6.93 kg/m³
Average kinetic flame temperature 923.21 K
Heat flux density due to metal combustion, 343,449.26 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 5.93e-08 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.74e-05 m
Pore diameter 9.73e-11 m
Diffusion flame heat flux density 243,034.01 W/m²
Diffusion flame height 1,223.93 μm
Heat flux density to surface within skeletal structure 439,255.15 W/m²
Heat flux density to surface outside skeletal structure 46,950.61 W/m²
Total heat flux density to surface 729,239.76 W/m²
Binder sublimation heat flux density 729,239.76 W/m²
Surface heat balance error -7.57e-05 W/m²

At pressure 2.22 MPa

Experimental burning rate 17.01 mm/s
Calculated burning rate 16.81 mm/s

First Conditional Fuel

Surface temperature 681.42 K
Linear burning rate 26.44 mm/s
Mass decomposition rate 43.87 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 9.12e+06 W/m²
Kinetic flame height 33.17 μm
Average kinetic flame density 4.02 kg/m³
Average kinetic flame temperature 2,194.21 K
Binder sublimation heat flux density 9.12e+06 W/m²
Surface heat balance error -0.0015 W/m²

Secondary Conditional Fuel

Surface temperature 611.04 K
Linear burning rate 5.4 mm/s
Mass decomposition rate 8.95 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 99,020.66 W/m²
Kinetic flame height 435.22 μm
Average kinetic flame density 10.67 kg/m³
Average kinetic flame temperature 826.52 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.07e+06 W/m²
Kinetic flame height 59.13 μm
Average kinetic flame density 9.52 kg/m³

Average kinetic flame temperature 926.02 K
 Heat flux density due to metal combustion, 429,629.02 W/m²
 Effective thermal conductivity 1.41 W/(m·K)
 Conductive thermal conductivity 1.41 W/(m·K)
 Radiative thermal conductivity 5.64e-08 W/(m·K)
 Conductive thermal conductivity balance error 9.240271880983641E-10
 Skeleton layer thickness 1.39e-05 m
 Pore diameter 9.25e-11 m
 Diffusion flame heat flux density 210,349.87 W/m²
 Diffusion flame height 1,411.44 μm
 Heat flux density to surface within skeletal structure 649,988.49 W/m²
 Heat flux density to surface outside skeletal structure 55,969.15 W/m²
 Total heat flux density to surface 916,307.51 W/m²
 Binder sublimation heat flux density 916,307.51 W/m²
 Surface heat balance error 2.47e-05 W/m²

At pressure 2.83 MPa

Experimental burning rate 20.17 mm/s
 Calculated burning rate 19.52 mm/s

First Conditional Fuel

Surface temperature 694.37 K
 Linear burning rate 34.2 mm/s
 Mass decomposition rate 56.74 kg·s⁻¹·m⁻²
 Kinetic flame heat flux density to surface 1.29e+07 W/m²
 Kinetic flame height 23.36 μm
 Average kinetic flame density 5.11 kg/m³
 Average kinetic flame temperature 2,200.69 K
 Binder sublimation heat flux density 1.29e+07 W/m²
 Surface heat balance error 0.0016 W/m²

Secondary Conditional Fuel

Surface temperature 616.36 K
 Linear burning rate 6.16 mm/s
 Mass decomposition rate 10.23 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 106,742.29 W/m²
 Kinetic flame height 398.75 μm
 Average kinetic flame density 13.56 kg/m³
 Average kinetic flame temperature 829.18 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.58e+06 W/m²
 Kinetic flame height 39.54 μm
 Average kinetic flame density 12.11 kg/m³
 Average kinetic flame temperature 928.68 K
 Heat flux density due to metal combustion, 529,304.99 W/m²
 Effective thermal conductivity 1.41 W/(m·K)
 Conductive thermal conductivity 1.41 W/(m·K)
 Radiative thermal conductivity 5.38e-08 W/(m·K)
 Conductive thermal conductivity balance error 9.240271880983641E-10
 Skeleton layer thickness 1.12e-05 m
 Pore diameter 8.83e-11 m
 Diffusion flame heat flux density 183,845.7 W/m²
 Diffusion flame height 1,612.03 μm
 Heat flux density to surface within skeletal structure 882,265.98 W/m²
 Heat flux density to surface outside skeletal structure 62,087.25 W/m²
 Total heat flux density to surface 1.13e+06 W/m²
 Binder sublimation heat flux density 1.13e+06 W/m²
 Surface heat balance error -2.38e-04 W/m²

At pressure 3.44 MPa

Experimental burning rate 23.12 mm/s
Calculated burning rate 22.25 mm/s

First Conditional Fuel

Surface temperature 705.21 K
Linear burning rate 42.11 mm/s
Mass decomposition rate $69.86 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.7\text{e}+07 \text{ W/m}^2$
Kinetic flame height 17.64 μm
Average kinetic flame density 6.2 kg/m^3
Average kinetic flame temperature 2,206.11 K
Binder sublimation heat flux density $1.7\text{e}+07 \text{ W/m}^2$
Surface heat balance error $2.25\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 621.28 K
Linear burning rate 6.96 mm/s
Mass decomposition rate $11.54 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 111,891.24 W/m^2
Kinetic flame height 376.01 μm
Average kinetic flame density 16.44 kg/m^3
Average kinetic flame temperature 831.64 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.14\text{e}+06 \text{ W/m}^2$
Kinetic flame height 28.98 μm
Average kinetic flame density 14.68 kg/m^3
Average kinetic flame temperature 931.14 K
Heat flux density due to metal combustion, 639,831.47 W/m^2
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $5.16\text{e}-08 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $9.25\text{e}-06 \text{ m}$
Pore diameter $8.46\text{e}-11 \text{ m}$

Diffusion flame heat flux density 162,653.19 W/m^2
Diffusion flame height 1,819.03 μm
Heat flux density to surface within skeletal structure $1.13\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure 66,417.69 W/m^2
Total heat flux density to surface $1.36\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $1.36\text{e}+06 \text{ W/m}^2$
Surface heat balance error $2.73\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 25 mm/s

First Conditional Fuel

Surface temperature 714.58 K
Linear burning rate 50.15 mm/s
Mass decomposition rate $83.2 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.14\text{e}+07 \text{ W/m}^2$
Kinetic flame height 13.96 μm
Average kinetic flame density 7.28 kg/m^3
Average kinetic flame temperature 2,210.79 K
Binder sublimation heat flux density $2.14\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-7.84\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 625.8 K
Linear burning rate 7.76 mm/s
Mass decomposition rate $12.87 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 115,470.95 W/m²
Kinetic flame height 360.43 μm
Average kinetic flame density 19.3 kg/m³
Average kinetic flame temperature 833.9 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.73e+06 W/m²
Kinetic flame height 22.51 μm
Average kinetic flame density 17.24 kg/m³
Average kinetic flame temperature 933.4 K
Heat flux density due to metal combustion, 759,568 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 4.96e-08 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 7.77e-06 m
Pore diameter 8.14e-11 m

Diffusion flame heat flux density 145,586.97 W/m²
Diffusion flame height 2,029.16 μm
Heat flux density to surface within skeletal structure 1.39e+06 W/m²
Heat flux density to surface outside skeletal structure 69,605.23 W/m²
Total heat flux density to surface 1.6e+06 W/m²
Binder sublimation heat flux density 1.6e+06 W/m²
Surface heat balance error -2.54e-05 W/m²

At pressure 4.67 MPa

Experimental burning rate 28.6 mm/s
Calculated burning rate 27.76 mm/s

First Conditional Fuel

Surface temperature 722.86 K
Linear burning rate 58.31 mm/s
Mass decomposition rate 96.74 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.61e+07 W/m²
Kinetic flame height 11.42 μm
Average kinetic flame density 8.36 kg/m³
Average kinetic flame temperature 2,214.93 K
Binder sublimation heat flux density 2.61e+07 W/m²
Surface heat balance error -0.0044 W/m²

Secondary Conditional Fuel

Surface temperature 629.96 K
Linear burning rate 8.57 mm/s
Mass decomposition rate 14.22 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 118,045.34 W/m²
Kinetic flame height 349.05 μm
Average kinetic flame density 22.16 kg/m³
Average kinetic flame temperature 835.98 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.36e+06 W/m²
Kinetic flame height 18.2 μm
Average kinetic flame density 19.8 kg/m³
Average kinetic flame temperature 935.48 K
Heat flux density due to metal combustion, 887,419.79 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 4.79e-08 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 6.64e-06 m
Pore diameter 7.86e-11 m

Diffusion flame heat flux density 131,657.37 W/m²

Diffusion flame height 2,240.69 μm
Heat flux density to surface within skeletal structure $1.65\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $72,029.79 \text{ W/m}^2$
Total heat flux density to surface $1.86\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $1.86\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-2.04\text{e}-04 \text{ W/m}^2$

At pressure 5.28 MPa

Experimental burning rate 31.17 mm/s
Calculated burning rate 30.52 mm/s

First Conditional Fuel

Surface temperature 730.3 K
Linear burning rate 66.57 mm/s
Mass decomposition rate $110.45 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $3.11\text{e}+07 \text{ W/m}^2$
Kinetic flame height 9.58 μm
Average kinetic flame density 9.44 kg/m^3
Average kinetic flame temperature 2,218.65 K
Binder sublimation heat flux density $3.11\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0019 W/m^2

Secondary Conditional Fuel

Surface temperature 633.8 K
Linear burning rate 9.38 mm/s
Mass decomposition rate $15.56 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $119,947.03 \text{ W/m}^2$
Kinetic flame height 340.31 μm
Average kinetic flame density 25 kg/m^3
Average kinetic flame temperature 837.9 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $4.01\text{e}+06 \text{ W/m}^2$
Kinetic flame height 15.16 μm
Average kinetic flame density 22.35 kg/m^3
Average kinetic flame temperature 937.4 K
Heat flux density due to metal combustion, $1.02\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity $1.41 \text{ W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity $4.64\text{e}-08 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $5.74\text{e}-06 \text{ m}$
Pore diameter $7.61\text{e}-11 \text{ m}$
Diffusion flame heat flux density $120,121.14 \text{ W/m}^2$
Diffusion flame height 2,452.69 μm
Heat flux density to surface within skeletal structure $1.93\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $73,924.74 \text{ W/m}^2$
Total heat flux density to surface $2.12\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $2.12\text{e}+06 \text{ W/m}^2$
Surface heat balance error $2.99\text{e}-04 \text{ W/m}^2$

At pressure 5.89 MPa

Experimental burning rate 33.65 mm/s
Calculated burning rate 33.27 mm/s

First Conditional Fuel

Surface temperature 737.07 K
Linear burning rate 74.93 mm/s
Mass decomposition rate $124.31 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $3.62\text{e}+07 \text{ W/m}^2$
Kinetic flame height 8.2 μm
Average kinetic flame density 10.52 kg/m^3

Average kinetic flame temperature 2,222.04 K
Binder sublimation heat flux density $3.62\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0054 W/m^2

Secondary Conditional Fuel

Surface temperature 637.37 K
Linear burning rate 10.19 mm/s
Mass decomposition rate $16.91 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $121,379.56 \text{ W/m}^2$
Kinetic flame height $333.36 \mu\text{m}$
Average kinetic flame density 27.84 kg/m^3
Average kinetic flame temperature 839.68 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $4.68\text{e}+06 \text{ W/m}^2$
Kinetic flame height $12.9 \mu\text{m}$
Average kinetic flame density 24.89 kg/m^3
Average kinetic flame temperature 939.18 K
Heat flux density due to metal combustion, $1.16\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $4.51\text{e}-08 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $5.03\text{e}-06 \text{ m}$
Pore diameter $7.39\text{e}-11 \text{ m}$
Diffusion flame heat flux density $110,431.69 \text{ W/m}^2$
Diffusion flame height $2,664.67 \mu\text{m}$
Heat flux density to surface within skeletal structure $2.21\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $75,436.55 \text{ W/m}^2$
Total heat flux density to surface $2.4\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $2.4\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-3.3\text{e}-05 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 36.02 mm/s

First Conditional Fuel

Surface temperature 743.3 K
Linear burning rate 83.37 mm/s
Mass decomposition rate $138.31 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $4.16\text{e}+07 \text{ W/m}^2$
Kinetic flame height $7.13 \mu\text{m}$
Average kinetic flame density 11.59 kg/m^3
Average kinetic flame temperature 2,225.15 K
Binder sublimation heat flux density $4.16\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0025 W/m^2

Secondary Conditional Fuel

Surface temperature 640.69 K
Linear burning rate 11 mm/s
Mass decomposition rate $18.25 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $122,476 \text{ W/m}^2$
Kinetic flame height $327.67 \mu\text{m}$
Average kinetic flame density 30.66 kg/m^3
Average kinetic flame temperature 841.34 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.37\text{e}+06 \text{ W/m}^2$
Kinetic flame height $11.18 \mu\text{m}$
Average kinetic flame density 27.42 kg/m^3
Average kinetic flame temperature 940.84 K

Heat flux density due to metal combustion, $1.31\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $4.39\text{e}-08 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $4.46\text{e}-06 \text{ m}$
Pore diameter $7.2\text{e}-11 \text{ m}$
Diffusion flame heat flux density $102,190.53 \text{ W/m}^2$
Diffusion flame height $2,876.31 \mu\text{m}$
Heat flux density to surface within skeletal structure $2.5\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $76,665.57 \text{ W/m}^2$
Total heat flux density to surface $2.68\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $2.68\text{e}+06 \text{ W/m}^2$
Surface heat balance error $4.61\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 14.46 mm/s

First Conditional Fuel

Surface temperature 628.08 K
Linear burning rate 12.52 mm/s
Mass decomposition rate $20.76 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.66\text{e}+06 \text{ W/m}^2$
Kinetic flame height $115.95 \mu\text{m}$
Average kinetic flame density 1.83 kg/m^3
Average kinetic flame temperature $2,167.54 \text{ K}$
Binder sublimation heat flux density $2.66\text{e}+06 \text{ W/m}^2$
Surface heat balance error $8.95\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 617.78 K
Linear burning rate 7.4 mm/s
Mass decomposition rate $12.28 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.4\text{e}+06 \text{ W/m}^2$
Kinetic flame height $72.92 \mu\text{m}$
Average kinetic flame density 2.66 kg/m^3
Average kinetic flame temperature $1,491.39 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $543,214.84 \text{ W/m}^2$
Kinetic flame height $179.16 \mu\text{m}$
Average kinetic flame density 3.59 kg/m^3
Average kinetic flame temperature $1,104.39 \text{ K}$
Heat flux density due to metal combustion, $23,962.66 \text{ W/m}^2$
Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0091 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $7.24\text{e}-05 \text{ m}$
Pore diameter $5.67\text{e}-06 \text{ m}$
Diffusion flame heat flux density $280,024.95 \text{ W/m}^2$
Diffusion flame height $1,088.55 \mu\text{m}$
Heat flux density to surface within skeletal structure $401,827.99 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $698,573.05 \text{ W/m}^2$
Total heat flux density to surface $1.38\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $1.38\text{e}+06 \text{ W/m}^2$
Surface heat balance error $4.43\text{e}-04 \text{ W/m}^2$

At pressure 1.61 MPa

Experimental burning rate 19.92 mm/s

Calculated burning rate 19.92 mm/s

First Conditional Fuel

Surface temperature 638.93 K

Linear burning rate 21.37 mm/s

Mass decomposition rate $35.45 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $5.11\text{e}+06 \text{ W/m}^2$

Kinetic flame height 60.04 μm

Average kinetic flame density 2.94 kg/m^3

Average kinetic flame temperature 2,172.96 K

Binder sublimation heat flux density $5.11\text{e}+06 \text{ W/m}^2$

Surface heat balance error $9.43\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 623.08 K

Linear burning rate 9.72 mm/s

Mass decomposition rate $16.13 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.71\text{e}+06 \text{ W/m}^2$

Kinetic flame height 64.35 μm

Average kinetic flame density 4.28 kg/m^3

Average kinetic flame temperature 1,494.04 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.12\text{e}+06 \text{ W/m}^2$

Kinetic flame height 86.24 μm

Average kinetic flame density 5.78 kg/m^3

Average kinetic flame temperature 1,107.04 K

Heat flux density due to metal combustion, 36,823 W/m^2

Effective thermal conductivity 0.41 $\text{W}/(\text{m}\cdot\text{K})$

Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$

Radiative thermal conductivity 0.0083 $\text{W}/(\text{m}\cdot\text{K})$

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness $4.69\text{e}-05 \text{ m}$

Pore diameter $5.15\text{e}-06 \text{ m}$

Diffusion flame heat flux density 212,797.2 W/m^2

Diffusion flame height 1,429.96 μm

Heat flux density to surface within skeletal structure 732,452.98 W/m^2

Heat flux density to surface outside skeletal structure 996,500.35 W/m^2

Total heat flux density to surface $1.94\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $1.94\text{e}+06 \text{ W/m}^2$

Surface heat balance error $-2.03\text{e}-04 \text{ W/m}^2$

At pressure 2.22 MPa

Experimental burning rate 24.95 mm/s

Calculated burning rate 24.86 mm/s

First Conditional Fuel

Surface temperature 646.5 K

Linear burning rate 30.72 mm/s

Mass decomposition rate $50.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $7.93\text{e}+06 \text{ W/m}^2$

Kinetic flame height 38.61 μm

Average kinetic flame density 4.05 kg/m^3

Average kinetic flame temperature 2,176.75 K

Binder sublimation heat flux density $7.93\text{e}+06 \text{ W/m}^2$

Surface heat balance error -0.0028 W/m^2

Secondary Conditional Fuel

Surface temperature 626.95 K

Linear burning rate 11.82 mm/s

Mass decomposition rate $19.62 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.9\text{e}+06 \text{ W/m}^2$

Kinetic flame height $59.85 \text{ }\mu\text{m}$

Average kinetic flame density 5.9 kg/m^3

Average kinetic flame temperature $1,495.97 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.81\text{e}+06 \text{ W/m}^2$

Kinetic flame height $53.18 \text{ }\mu\text{m}$

Average kinetic flame density 7.95 kg/m^3

Average kinetic flame temperature $1,108.97 \text{ K}$

Heat flux density due to metal combustion, $50,128.46 \text{ W/m}^2$

Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.0077 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$

Skeleton layer thickness $3.43\text{e-}05 \text{ m}$

Pore diameter $4.81\text{e-}06 \text{ m}$

Diffusion flame heat flux density $174,731.46 \text{ W/m}^2$

Diffusion flame height $1,739.27 \text{ }\mu\text{m}$

Heat flux density to surface within skeletal structure $1.08\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $1.22\text{e}+06 \text{ W/m}^2$

Total heat flux density to surface $2.48\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $2.48\text{e}+06 \text{ W/m}^2$

Surface heat balance error $-6.44\text{e-}04 \text{ W/m}^2$

At pressure 2.83 MPa

Experimental burning rate 29.57 mm/s

Calculated burning rate 29.43 mm/s

First Conditional Fuel

Surface temperature 652.37 K

Linear burning rate 40.46 mm/s

Mass decomposition rate $67.12 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $1.1\text{e}+07 \text{ W/m}^2$

Kinetic flame height $27.7 \text{ }\mu\text{m}$

Average kinetic flame density 5.16 kg/m^3

Average kinetic flame temperature $2,179.69 \text{ K}$

Binder sublimation heat flux density $1.1\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0035 W/m^2

Secondary Conditional Fuel

Surface temperature 629.99 K

Linear burning rate 13.77 mm/s

Mass decomposition rate $22.85 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.05\text{e}+06 \text{ W/m}^2$

Kinetic flame height $56.9 \text{ }\mu\text{m}$

Average kinetic flame density 7.51 kg/m^3

Average kinetic flame temperature $1,497.5 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.59\text{e}+06 \text{ W/m}^2$

Kinetic flame height $37.05 \text{ }\mu\text{m}$

Average kinetic flame density 10.13 kg/m^3

Average kinetic flame temperature $1,110.5 \text{ K}$

Heat flux density due to metal combustion, $63,737.93 \text{ W/m}^2$

Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.0073 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$

Skeleton layer thickness $2.69\text{e-}05 \text{ m}$

Pore diameter $4.56\text{e-}06 \text{ m}$

Diffusion flame heat flux density 149,874.19 W/m²
Diffusion flame height 2,025.71 μ m
Heat flux density to surface within skeletal structure 1.44e+06 W/m²
Heat flux density to surface outside skeletal structure 1.4e+06 W/m²
Total heat flux density to surface 2.99e+06 W/m²
Binder sublimation heat flux density 2.99e+06 W/m²
Surface heat balance error -0.0014 W/m²

At pressure 3.44 MPa

Experimental burning rate 33.9 mm/s
Calculated burning rate 33.75 mm/s

First Conditional Fuel

Surface temperature 657.18 K
Linear burning rate 50.51 mm/s
Mass decomposition rate 83.8 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.44e+07 W/m²
Kinetic flame height 21.21 μ m
Average kinetic flame density 6.27 kg/m³
Average kinetic flame temperature 2,182.09 K
Binder sublimation heat flux density 1.44e+07 W/m²
Surface heat balance error 0.0043 W/m²

Secondary Conditional Fuel

Surface temperature 632.51 K
Linear burning rate 15.61 mm/s
Mass decomposition rate 25.89 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.16e+06 W/m²
Kinetic flame height 54.77 μ m
Average kinetic flame density 9.12 kg/m³
Average kinetic flame temperature 1,498.75 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.45e+06 W/m²
Kinetic flame height 27.76 μ m
Average kinetic flame density 12.3 kg/m³
Average kinetic flame temperature 1,111.75 K
Heat flux density due to metal combustion, 77,611.78 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.007 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.21e-05 m
Pore diameter 4.36e-06 m

Diffusion flame heat flux density 132,155.72 W/m²
Diffusion flame height 2,295.39 μ m
Heat flux density to surface within skeletal structure 1.8e+06 W/m²
Heat flux density to surface outside skeletal structure 1.55e+06 W/m²
Total heat flux density to surface 3.48e+06 W/m²
Binder sublimation heat flux density 3.48e+06 W/m²
Surface heat balance error -0.0016 W/m²

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 37.87 mm/s

First Conditional Fuel

Surface temperature 661.27 K
Linear burning rate 60.84 mm/s
Mass decomposition rate 100.94 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.79e+07 W/m²
Kinetic flame height 16.98 μ m

Average kinetic flame density 7.37 kg/m³
Average kinetic flame temperature 2,184.13 K
Binder sublimation heat flux density 1.79e+07 W/m²
Surface heat balance error -0.0025 W/m²

Secondary Conditional Fuel

Surface temperature 634.66 K
Linear burning rate 17.35 mm/s
Mass decomposition rate 28.79 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.26e+06 W/m²
Kinetic flame height 53.12 μm
Average kinetic flame density 10.73 kg/m³
Average kinetic flame temperature 1,499.83 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.38e+06 W/m²
Kinetic flame height 21.83 μm
Average kinetic flame density 14.46 kg/m³
Average kinetic flame temperature 1,112.83 K
Heat flux density due to metal combustion, 91,735.44 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0068 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.86e-05 m
Pore diameter 4.2e-06 m

Diffusion flame heat flux density 118,768.33 W/m²
Diffusion flame height 2,552.31 μm
Heat flux density to surface within skeletal structure 2.17e+06 W/m²
Heat flux density to surface outside skeletal structure 1.68e+06 W/m²
Total heat flux density to surface 3.97e+06 W/m²
Binder sublimation heat flux density 3.97e+06 W/m²
Surface heat balance error -0.0016 W/m²

At pressure 4.67 MPa

Experimental burning rate 41.93 mm/s
Calculated burning rate 41.83 mm/s

First Conditional Fuel

Surface temperature 664.83 K
Linear burning rate 71.41 mm/s
Mass decomposition rate 118.47 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.17e+07 W/m²
Kinetic flame height 14.03 μm
Average kinetic flame density 8.47 kg/m³
Average kinetic flame temperature 2,185.91 K
Binder sublimation heat flux density 2.17e+07 W/m²
Surface heat balance error -0.0091 W/m²

Secondary Conditional Fuel

Surface temperature 636.54 K
Linear burning rate 19.03 mm/s
Mass decomposition rate 31.57 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.34e+06 W/m²
Kinetic flame height 51.81 μm
Average kinetic flame density 12.34 kg/m³
Average kinetic flame temperature 1,500.77 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.37e+06 W/m²
Kinetic flame height 17.77 μm
Average kinetic flame density 16.63 kg/m³

Average kinetic flame temperature 1,113.77 K
Heat flux density due to metal combustion, 106,101.71 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0065 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.61e-05 m
Pore diameter 4.07e-06 m
Diffusion flame heat flux density 108,227.27 W/m²
Diffusion flame height 2,799.16 μm
Heat flux density to surface within skeletal structure 2.54e+06 W/m²
Heat flux density to surface outside skeletal structure 1.79e+06 W/m²
Total heat flux density to surface 4.44e+06 W/m²
Binder sublimation heat flux density 4.44e+06 W/m²
Surface heat balance error 9.52e-04 W/m²

At pressure 5.28 MPa

Experimental burning rate 45.71 mm/s
Calculated burning rate 45.66 mm/s

First Conditional Fuel

Surface temperature 667.98 K
Linear burning rate 82.19 mm/s
Mass decomposition rate 136.36 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.56e+07 W/m²
Kinetic flame height 11.87 μm
Average kinetic flame density 9.58 kg/m³
Average kinetic flame temperature 2,187.49 K
Binder sublimation heat flux density 2.56e+07 W/m²
Surface heat balance error 0.0098 W/m²

Secondary Conditional Fuel

Surface temperature 638.22 K
Linear burning rate 20.65 mm/s
Mass decomposition rate 34.26 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.4e+06 W/m²
Kinetic flame height 50.73 μm
Average kinetic flame density 13.95 kg/m³
Average kinetic flame temperature 1,501.61 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.41e+06 W/m²
Kinetic flame height 14.85 μm
Average kinetic flame density 18.79 kg/m³
Average kinetic flame temperature 1,114.61 K
Heat flux density due to metal combustion, 120,705.89 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0063 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.41e-05 m
Pore diameter 3.95e-06 m
Diffusion flame heat flux density 99,668.85 W/m²
Diffusion flame height 3,037.84 μm
Heat flux density to surface within skeletal structure 2.92e+06 W/m²
Heat flux density to surface outside skeletal structure 1.88e+06 W/m²
Total heat flux density to surface 4.9e+06 W/m²
Binder sublimation heat flux density 4.9e+06 W/m²
Surface heat balance error -9.37e-04 W/m²

At pressure 5.89 MPa

Experimental burning rate 49.35 mm/s
Calculated burning rate 49.39 mm/s

First Conditional Fuel

Surface temperature 670.82 K
Linear burning rate 93.17 mm/s
Mass decomposition rate $154.57 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.97\text{e}+07 \text{ W/m}^2$
Kinetic flame height 10.23 μm
Average kinetic flame density 10.68 kg/m^3
Average kinetic flame temperature 2,188.91 K
Binder sublimation heat flux density $2.97\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0085 W/m^2

Secondary Conditional Fuel

Surface temperature 639.74 K
Linear burning rate 22.23 mm/s
Mass decomposition rate $36.88 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.46\text{e}+06 \text{ W/m}^2$
Kinetic flame height 49.82 μm
Average kinetic flame density 15.56 kg/m^3
Average kinetic flame temperature 1,502.37 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $7.51\text{e}+06 \text{ W/m}^2$
Kinetic flame height 12.67 μm
Average kinetic flame density 20.96 kg/m^3
Average kinetic flame temperature 1,115.37 K
Heat flux density due to metal combustion, 135,544.1 W/m^2
Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0062 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.25\text{e}-05 \text{ m}$
Pore diameter $3.85\text{e}-06 \text{ m}$

Diffusion flame heat flux density $92,553.81 \text{ W/m}^2$
Diffusion flame height 3,269.73 μm
Heat flux density to surface within skeletal structure $3.3\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.97\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $5.36\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $5.36\text{e}+06 \text{ W/m}^2$
Surface heat balance error -0.002 W/m^2

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
Calculated burning rate 53.03 mm/s

First Conditional Fuel

Surface temperature 673.4 K
Linear burning rate 104.33 mm/s
Mass decomposition rate $173.08 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $3.39\text{e}+07 \text{ W/m}^2$
Kinetic flame height 8.95 μm
Average kinetic flame density 11.78 kg/m^3
Average kinetic flame temperature 2,190.2 K
Binder sublimation heat flux density $3.39\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0095 W/m^2

Secondary Conditional Fuel

Surface temperature 641.13 K
Linear burning rate 23.77 mm/s
Mass decomposition rate $39.43 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.52 \times 10^6 \text{ W/m}^2$
Kinetic flame height $49.04 \text{ }\mu\text{m}$
Average kinetic flame density 17.16 kg/m^3
Average kinetic flame temperature $1,503.06 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.65 \times 10^6 \text{ W/m}^2$
Kinetic flame height $10.98 \text{ }\mu\text{m}$
Average kinetic flame density 23.12 kg/m^3
Average kinetic flame temperature $1,116.06 \text{ K}$
Heat flux density due to metal combustion, $150,612.3 \text{ W/m}^2$
Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.006 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $1.13 \times 10^{-5} \text{ m}$
Pore diameter $3.76 \times 10^{-6} \text{ m}$

Diffusion flame heat flux density $86,526.7 \text{ W/m}^2$
Diffusion flame height $3,495.88 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $3.68 \times 10^6 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $2.04 \times 10^6 \text{ W/m}^2$
Total heat flux density to surface $5.81 \times 10^6 \text{ W/m}^2$
Binder sublimation heat flux density $5.81 \times 10^6 \text{ W/m}^2$
Surface heat balance error -0.0018 W/m^2

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s
Calculated burning rate 5.98 mm/s

First Conditional Fuel

Surface temperature 837.49 K
Linear burning rate 6.53 mm/s
Mass decomposition rate $10.84 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $4.79 \times 10^6 \text{ W/m}^2$
Kinetic flame height $59.91 \text{ }\mu\text{m}$
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature $2,272.24 \text{ K}$
Binder sublimation heat flux density $4.79 \times 10^6 \text{ W/m}^2$
Surface heat balance error $-5.47 \times 10^{-4} \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 816.95 K
Linear burning rate 2.91 mm/s
Mass decomposition rate $4.83 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $6.59 \times 10^6 \text{ W/m}^2$
Kinetic flame height $23.49 \text{ }\mu\text{m}$
Average kinetic flame density 2.49 kg/m^3
Average kinetic flame temperature $1,590.97 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $168,201.77 \text{ W/m}^2$
Kinetic flame height $460.19 \text{ }\mu\text{m}$
Average kinetic flame density 3.3 kg/m^3
Average kinetic flame temperature $1,203.97 \text{ K}$
Heat flux density due to metal combustion, $65,908.79 \text{ W/m}^2$
Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0036 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $2.29 \times 10^{-5} \text{ m}$

Pore diameter 2.27e-06 m
Diffusion flame heat flux density 665,715.83 W/m²
Diffusion flame height 427.97 μm
Heat flux density to surface within skeletal structure 194,153.18 W/m²
Heat flux density to surface outside skeletal structure 1.12e+06 W/m²
Total heat flux density to surface 1.98e+06 W/m²
Binder sublimation heat flux density 1.98e+06 W/m²
Surface heat balance error 5.27e-04 W/m²

At pressure 1.61 MPa

Experimental burning rate 7.9 mm/s
Calculated burning rate 7.86 mm/s

First Conditional Fuel

Surface temperature 852.57 K
Linear burning rate 11.54 mm/s
Mass decomposition rate 19.14 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 8.89e+06 W/m²
Kinetic flame height 32.1 μm
Average kinetic flame density 2.8 kg/m³
Average kinetic flame temperature 2,279.79 K
Binder sublimation heat flux density 8.89e+06 W/m²
Surface heat balance error 1.12e-04 W/m²

Secondary Conditional Fuel

Surface temperature 822.58 K
Linear burning rate 3.65 mm/s
Mass decomposition rate 6.05 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 7.8e+06 W/m²
Kinetic flame height 19.79 μm
Average kinetic flame density 4.01 kg/m³
Average kinetic flame temperature 1,593.79 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 156,584.41 W/m²
Kinetic flame height 490.74 μm
Average kinetic flame density 5.3 kg/m³
Average kinetic flame temperature 1,206.79 K
Heat flux density due to metal combustion, 94,031.89 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0034 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.6e-05 m
Pore diameter 2.09e-06 m
Diffusion flame heat flux density 530,126.74 W/m²
Diffusion flame height 536.37 μm
Heat flux density to surface within skeletal structure 192,222.55 W/m²
Heat flux density to surface outside skeletal structure 1.82e+06 W/m²
Total heat flux density to surface 2.54e+06 W/m²
Binder sublimation heat flux density 2.54e+06 W/m²
Surface heat balance error 9.9e-05 W/m²

At pressure 2.22 MPa

Experimental burning rate 9.56 mm/s
Calculated burning rate 9.53 mm/s

First Conditional Fuel

Surface temperature 863.05 K
Linear burning rate 16.93 mm/s
Mass decomposition rate 28.09 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.35e+07 W/m²

Kinetic flame height 21.08 μm
Average kinetic flame density 3.86 kg/m^3
Average kinetic flame temperature 2,285.03 K
Binder sublimation heat flux density 1.35e+07 W/m^2
Surface heat balance error -5.08e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 826.84 K
Linear burning rate 4.32 mm/s
Mass decomposition rate 7.16 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 8.59e+06 W/m^2
Kinetic flame height 17.91 μm
Average kinetic flame density 5.53 kg/m^3
Average kinetic flame temperature 1,595.92 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 146,876.27 W/m^2
Kinetic flame height 520.27 μm
Average kinetic flame density 7.3 kg/m^3
Average kinetic flame temperature 1,208.92 K
Heat flux density due to metal combustion, 122,635.54 W/m^2
Effective thermal conductivity 0.41 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 0.0032 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.22e-05 m
Pore diameter 1.97e-06 m
Diffusion flame heat flux density 447,144.02 W/m^2
Diffusion flame height 634.95 μm
Heat flux density to surface within skeletal structure 193,872.07 W/m^2
Heat flux density to surface outside skeletal structure 2.41e+06 W/m^2
Total heat flux density to surface 3.05e+06 W/m^2
Binder sublimation heat flux density 3.05e+06 W/m^2
Surface heat balance error -6.45e-04 W/m^2

At pressure 2.83 MPa

Experimental burning rate 11.03 mm/s
Calculated burning rate 11.03 mm/s

First Conditional Fuel

Surface temperature 871.15 K
Linear burning rate 22.62 mm/s
Mass decomposition rate 37.53 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 1.85e+07 W/m^2
Kinetic flame height 15.34 μm
Average kinetic flame density 4.91 kg/m^3
Average kinetic flame temperature 2,289.07 K
Binder sublimation heat flux density 1.85e+07 W/m^2
Surface heat balance error 0.0012 W/m^2

Secondary Conditional Fuel

Surface temperature 830.2 K
Linear burning rate 4.93 mm/s
Mass decomposition rate 8.17 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 9.2e+06 W/m^2
Kinetic flame height 16.69 μm
Average kinetic flame density 7.04 kg/m^3
Average kinetic flame temperature 1,597.6 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 139,360.48 W/m^2
Kinetic flame height 545.92 μm

Average kinetic flame density 9.29 kg/m³
Average kinetic flame temperature 1,210.6 K
Heat flux density due to metal combustion, 150,917.35 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.003 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 9.9e-06 m
Pore diameter 1.88e-06 m
Diffusion flame heat flux density 391,446.37 W/m²
Diffusion flame height 724.44 μm
Heat flux density to surface within skeletal structure 197,716.48 W/m²
Heat flux density to surface outside skeletal structure 2.93e+06 W/m²
Total heat flux density to surface 3.52e+06 W/m²
Binder sublimation heat flux density 3.52e+06 W/m²
Surface heat balance error -0.0011 W/m²

At pressure 3.44 MPa

Experimental burning rate 12.38 mm/s
Calculated burning rate 12.4 mm/s

First Conditional Fuel

Surface temperature 877.76 K
Linear burning rate 28.56 mm/s
Mass decomposition rate 47.38 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.38e+07 W/m²
Kinetic flame height 11.88 μm
Average kinetic flame density 5.96 kg/m³
Average kinetic flame temperature 2,292.38 K
Binder sublimation heat flux density 2.38e+07 W/m²
Surface heat balance error -0.0059 W/m²

Secondary Conditional Fuel

Surface temperature 832.95 K
Linear burning rate 5.48 mm/s
Mass decomposition rate 9.09 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 9.71e+06 W/m²
Kinetic flame height 15.78 μm
Average kinetic flame density 8.55 kg/m³
Average kinetic flame temperature 1,598.97 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 133,479.12 W/m²
Kinetic flame height 567.92 μm
Average kinetic flame density 11.28 kg/m³
Average kinetic flame temperature 1,211.97 K
Heat flux density due to metal combustion, 178,577.47 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0029 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 8.35e-06 m
Pore diameter 1.81e-06 m
Diffusion flame heat flux density 351,408.96 W/m²
Diffusion flame height 806.2 μm
Heat flux density to surface within skeletal structure 202,711.52 W/m²
Heat flux density to surface outside skeletal structure 3.4e+06 W/m²
Total heat flux density to surface 3.96e+06 W/m²
Binder sublimation heat flux density 3.96e+06 W/m²
Surface heat balance error 0.0011 W/m²

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s

Calculated burning rate 13.67 mm/s

First Conditional Fuel

Surface temperature 883.38 K

Linear burning rate 34.71 mm/s

Mass decomposition rate $57.58 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.94\text{e}+07 \text{ W/m}^2$

Kinetic flame height 9.6 μm

Average kinetic flame density 7.01 kg/m^3

Average kinetic flame temperature 2,295.19 K

Binder sublimation heat flux density $2.94\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0033 W/m^2

Secondary Conditional Fuel

Surface temperature 835.26 K

Linear burning rate 5.99 mm/s

Mass decomposition rate $9.94 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.02\text{e}+07 \text{ W/m}^2$

Kinetic flame height 15.05 μm

Average kinetic flame density 10.06 kg/m^3

Average kinetic flame temperature 1,600.13 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $128,756.16 \text{ W/m}^2$

Kinetic flame height 586.96 μm

Average kinetic flame density 13.27 kg/m^3

Average kinetic flame temperature 1,213.13 K

Heat flux density due to metal combustion, $205,550.9 \text{ W/m}^2$

Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.0028 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$

Skeleton layer thickness $7.24\text{e-}06 \text{ m}$

Pore diameter $1.76\text{e-}06 \text{ m}$

Diffusion flame heat flux density $321,099.81 \text{ W/m}^2$

Diffusion flame height 881.58 μm

Heat flux density to surface within skeletal structure $208,255.2 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $3.83\text{e}+06 \text{ W/m}^2$

Total heat flux density to surface $4.36\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $4.36\text{e}+06 \text{ W/m}^2$

Surface heat balance error $8.96\text{e-}04 \text{ W/m}^2$

At pressure 4.67 MPa

Experimental burning rate 14.8 mm/s

Calculated burning rate 14.84 mm/s

First Conditional Fuel

Surface temperature 888.26 K

Linear burning rate 41.04 mm/s

Mass decomposition rate $68.08 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $3.53\text{e}+07 \text{ W/m}^2$

Kinetic flame height 7.99 μm

Average kinetic flame density 8.06 kg/m^3

Average kinetic flame temperature 2,297.63 K

Binder sublimation heat flux density $3.53\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0011 W/m^2

Secondary Conditional Fuel

Surface temperature 837.24 K

Linear burning rate 6.47 mm/s

Mass decomposition rate $10.73 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.06 \times 10^7 \text{ W/m}^2$
Kinetic flame height $14.45 \text{ }\mu\text{m}$
Average kinetic flame density 11.57 kg/m^3
Average kinetic flame temperature $1,601.12 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $124,871.61 \text{ W/m}^2$
Kinetic flame height $603.63 \text{ }\mu\text{m}$
Average kinetic flame density 15.26 kg/m^3
Average kinetic flame temperature $1,214.12 \text{ K}$
Heat flux density due to metal combustion, $231,839.11 \text{ W/m}^2$
Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0027 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $6.41 \times 10^{-6} \text{ m}$
Pore diameter $1.71 \times 10^{-6} \text{ m}$
Diffusion flame heat flux density $297,251.03 \text{ W/m}^2$
Diffusion flame height $951.64 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $214,025.13 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $4.23 \times 10^6 \text{ W/m}^2$
Total heat flux density to surface $4.74 \times 10^6 \text{ W/m}^2$
Binder sublimation heat flux density $4.74 \times 10^6 \text{ W/m}^2$
Surface heat balance error $6.05 \times 10^{-5} \text{ W/m}^2$

At pressure 5.28 MPa

Experimental burning rate 15.92 mm/s
Calculated burning rate 15.94 mm/s

First Conditional Fuel

Surface temperature 892.58 K
Linear burning rate 47.53 mm/s
Mass decomposition rate $78.86 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $4.14 \times 10^7 \text{ W/m}^2$
Kinetic flame height $6.8 \text{ }\mu\text{m}$
Average kinetic flame density 9.11 kg/m^3
Average kinetic flame temperature $2,299.79 \text{ K}$
Binder sublimation heat flux density $4.14 \times 10^7 \text{ W/m}^2$
Surface heat balance error 0.0014 W/m^2

Secondary Conditional Fuel

Surface temperature 838.98 K
Linear burning rate 6.92 mm/s
Mass decomposition rate $11.47 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.09 \times 10^7 \text{ W/m}^2$
Kinetic flame height $13.94 \text{ }\mu\text{m}$
Average kinetic flame density 13.08 kg/m^3
Average kinetic flame temperature $1,601.99 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $121,609.55 \text{ W/m}^2$
Kinetic flame height $618.39 \text{ }\mu\text{m}$
Average kinetic flame density 17.24 kg/m^3
Average kinetic flame temperature $1,214.99 \text{ K}$
Heat flux density due to metal combustion, $257,473.72 \text{ W/m}^2$
Effective thermal conductivity $0.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0027 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $5.76 \times 10^{-6} \text{ m}$
Pore diameter $1.67 \times 10^{-6} \text{ m}$

Diffusion flame heat flux density 277,916.6 W/m²
Diffusion flame height 1,017.22 μm
Heat flux density to surface within skeletal structure 219,843.84 W/m²
Heat flux density to surface outside skeletal structure 4.6e+06 W/m²
Total heat flux density to surface 5.1e+06 W/m²
Binder sublimation heat flux density 5.1e+06 W/m²
Surface heat balance error 0.0016 W/m²

At pressure 5.89 MPa

Experimental burning rate 16.98 mm/s
Calculated burning rate 16.98 mm/s

First Conditional Fuel

Surface temperature 896.47 K
Linear burning rate 54.17 mm/s
Mass decomposition rate 89.87 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 4.77e+07 W/m²
Kinetic flame height 5.9 μm
Average kinetic flame density 10.15 kg/m³
Average kinetic flame temperature 2,301.73 K
Binder sublimation heat flux density 4.77e+07 W/m²
Surface heat balance error 0.0011 W/m²

Secondary Conditional Fuel

Surface temperature 840.52 K
Linear burning rate 7.34 mm/s
Mass decomposition rate 12.17 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.13e+07 W/m²
Kinetic flame height 13.49 μm
Average kinetic flame density 14.58 kg/m³
Average kinetic flame temperature 1,602.76 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 118,824.67 W/m²
Kinetic flame height 631.58 μm
Average kinetic flame density 19.22 kg/m³
Average kinetic flame temperature 1,215.76 K
Heat flux density due to metal combustion, 282,484.81 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0026 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.25e-06 m
Pore diameter 1.64e-06 m
Diffusion flame heat flux density 261,874.11 W/m²
Diffusion flame height 1,078.94 μm
Heat flux density to surface within skeletal structure 225,618.28 W/m²
Heat flux density to surface outside skeletal structure 4.95e+06 W/m²
Total heat flux density to surface 5.43e+06 W/m²
Binder sublimation heat flux density 5.43e+06 W/m²
Surface heat balance error -6.16e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 18 mm/s
Calculated burning rate 17.96 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 60.95 mm/s
Mass decomposition rate 101.12 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 5.42e+07 W/m²
Kinetic flame height 5.18 μm

Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 5.42e+07 W/m²
Surface heat balance error -0.015 W/m²

Secondary Conditional Fuel

Surface temperature 841.91 K
Linear burning rate 7.73 mm/s
Mass decomposition rate 12.83 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.16e+07 W/m²
Kinetic flame height 13.1 μm
Average kinetic flame density 16.09 kg/m³
Average kinetic flame temperature 1,603.45 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 116,410.87 W/m²
Kinetic flame height 643.49 μm
Average kinetic flame density 21.21 kg/m³
Average kinetic flame temperature 1,216.45 K
Heat flux density due to metal combustion, 306,919.04 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0026 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.82e-06 m
Pore diameter 1.61e-06 m

Diffusion flame heat flux density 248,304.38 W/m²
Diffusion flame height 1,137.35 μm
Heat flux density to surface within skeletal structure 231,292.07 W/m²
Heat flux density to surface outside skeletal structure 5.28e+06 W/m²
Total heat flux density to surface 5.75e+06 W/m²
Binder sublimation heat flux density 5.75e+06 W/m²
Surface heat balance error -0.0014 W/m²

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	13.94 mm/s		15.4 mm/s		11.51 mm/s		14.46 mm/s		5.98 mm/s	
Difference	3.01 mm/s		-0.16 mm/s		-1.78 mm/s		-0.19 mm/s		-0.017 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.54 mm/s	7.22 mm/s	11.54 mm/s	11.54 mm/s	11.54 mm/s	4.07 mm/s	12.52 mm/s	7.4 mm/s	6.53 mm/s	2.91 mm/s
Surface Temperature	642.77 K	622.79 K	642.77 K	642.77 K	642.77 K	600 K	628.08 K	617.78 K	837.49 K	816.95 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.87e+06 W/m ²		2.87e+06 W/m ²		2.87e+06 W/m ²		2.66e+06 W/m ²		4.79e+06 W/m ²	
Kinetic Flame Height (Inter-Pocket)	106.78 μm		106.78 μm		106.78 μm		115.95 μm		59.91 μm	
Kinetic Flame Heat Flux Density (Skeleton)	432,409.51 W/m ²		1.38e+06 W/m ²		251,404.51 W/m ²		543,214.84 W/m ²		168,201.77 W/m ²	
Kinetic Flame Height (Skeleton)	223.91 μm		143.6 μm		254.97 μm		179.16 μm		460.19 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.46e+06 W/m ²		1.56e+06 W/m ²		66,946.74 W/m ²		2.4e+06 W/m ²		6.59e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	70.72 μm		110.68 μm		660.23 μm		72.92 μm		23.49 μm	
Diffusion Flame Heat Flux Density	280,851.6 W/m ²		1.3e+06 W/m ²		280,194.16 W/m ²		280,024.95 W/m ²		665,715.83 W/m ²	
Diffusion Flame Height	1,061.49 μm		232.78 μm		1,063.55 μm		1,088.55 μm		427.97 μm	
Metal Combustion Heat Flux Density	193,963.72 W/m ²		222,442.1 W/m ²		275,437.08 W/m ²		23,962.66 W/m ²		65,908.79 W/m ²	

1.61 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	21.71 mm/s		20.68 mm/s		13.58 mm/s		19.92 mm/s		7.9 mm/s	
Calculated Burning Rate	18.57 mm/s		20.38 mm/s		14.16 mm/s		19.92 mm/s		7.86 mm/s	
Difference	3.14 mm/s		0.3 mm/s		-0.57 mm/s		-0.0045 mm/s		0.04 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	18.87 mm/s	9.16 mm/s	18.87 mm/s	13.35 mm/s	18.87 mm/s	4.68 mm/s	21.37 mm/s	9.72 mm/s	11.54 mm/s	3.65 mm/s
Surface Temperature	665.16 K	632.81 K	665.16 K	649.24 K	665.16 K	605.43 K	638.93 K	623.08 K	852.57 K	822.58 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	5.75e+06 W/m²		5.75e+06 W/m²		5.75e+06 W/m²		5.11e+06 W/m²		8.89e+06 W/m²	
Kinetic Flame Height (Inter-Pocket)	52.95 µm		52.95 µm		52.95 µm		60.04 µm		32.1 µm	
Kinetic Flame Heat Flux Density (Skeleton)	954,529.57 W/m²		3.33e+06 W/m²		612,328.97 W/m²		1.12e+06 W/m²		156,584.41 W/m²	
Kinetic Flame Height (Skeleton)	100.38 µm		59.34 µm		103.8 µm		86.24 µm		490.74 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.89e+06 W/m²		2e+06 W/m²		86,877.76 W/m²		2.71e+06 W/m²		7.8e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	60.03 µm		85.92 µm		502.51 µm		64.35 µm		19.79 µm	
Diffusion Flame Heat Flux Density	220,446.2 W/m²		1.12e+06 W/m²		243,034.01 W/m²		212,797.2 W/m²		530,126.74 W/m²	
Diffusion Flame Height	1,347.81 µm		269.27 µm		1,223.93 µm		1,429.96 µm		536.37 µm	
Metal Combustion Heat Flux Density	282,117.41 W/m²		279,462.85 W/m²		343,449.26 W/m²		36,823 W/m²		94,031.89 W/m²	

2.22 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	25.67 mm/s		25.41 mm/s		17.01 mm/s		24.95 mm/s		9.56 mm/s	
Calculated Burning Rate	22.71 mm/s		24.88 mm/s		16.81 mm/s		24.86 mm/s		9.53 mm/s	
Difference	2.95 mm/s		0.53 mm/s		0.2 mm/s		0.09 mm/s		0.026 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	26.44 mm/s	10.92 mm/s	26.44 mm/s	15.24 mm/s	26.44 mm/s	5.4 mm/s	30.72 mm/s	11.82 mm/s	16.93 mm/s	4.32 mm/s
Surface Temperature	681.42 K	640.35 K	681.42 K	655.25 K	681.42 K	611.04 K	646.5 K	626.95 K	863.05 K	826.84 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	9.12e+06 W/m ²		9.12e+06 W/m ²		9.12e+06 W/m ²		7.93e+06 W/m ²		1.35e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	33.17 μm		33.17 μm		33.17 μm		38.61 μm		21.08 μm	
Kinetic Flame Heat Flux Density (Skeleton)	1.61e+06 W/m ²		5.82e+06 W/m ²		1.07e+06 W/m ²		1.81e+06 W/m ²		146,876.27 W/m ²	
Kinetic Flame Height (Skeleton)	59.21 μm		33.83 μm		59.13 μm		53.18 μm		520.27 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.17e+06 W/m ²		2.28e+06 W/m ²		99,020.66 W/m ²		2.9e+06 W/m ²		8.59e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	54.47 μm		74.86 μm		435.22 μm		59.85 μm		17.91 μm	
Diffusion Flame Heat Flux Density	184,572.1 W/m ²		981,626.67 W/m ²		210,349.87 W/m ²		174,731.46 W/m ²		447,144.02 W/m ²	
Diffusion Flame Height	1,605.68 μm		307.53 μm		1,411.44 μm		1,739.27 μm		634.95 μm	
Metal Combustion Heat Flux Density	371,234.09 W/m ²		344,111.05 W/m ²		429,629.02 W/m ²		50,128.46 W/m ²		122,635.54 W/m ²	

2.83 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	29.12 mm/s		29.68 mm/s		20.17 mm/s		29.57 mm/s		11.03 mm/s	
Calculated Burning Rate	26.53 mm/s		29.19 mm/s		19.52 mm/s		29.43 mm/s		11.03 mm/s	
Difference	2.59 mm/s		0.5 mm/s		0.65 mm/s		0.14 mm/s		-0.0032 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	34.2 mm/s	12.53 mm/s	34.2 mm/s	17.18 mm/s	34.2 mm/s	6.16 mm/s	40.46 mm/s	13.77 mm/s	22.62 mm/s	4.93 mm/s
Surface Temperature	694.37 K	646.44 K	694.37 K	660.78 K	694.37 K	616.36 K	652.37 K	629.99 K	871.15 K	830.2 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.29e+07 W/m²		1.29e+07 W/m²		1.29e+07 W/m²		1.1e+07 W/m²		1.85e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	23.36 μm		23.36 μm		23.36 μm		27.7 μm		15.34 μm	
Kinetic Flame Heat Flux Density (Skeleton)	2.36e+06 W/m²		8.7e+06 W/m²		1.58e+06 W/m²		2.59e+06 W/m²		139,360.48 W/m²	
Kinetic Flame Height (Skeleton)	39.95 μm		22.56 μm		39.54 μm		37.05 μm		545.92 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.38e+06 W/m²		2.48e+06 W/m²		106,742.29 W/m²		3.05e+06 W/m²		9.2e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	50.89 μm		68.73 μm		398.75 μm		56.9 μm		16.69 μm	
Diffusion Flame Heat Flux Density	160,409.55 W/m²		869,013.86 W/m²		183,845.7 W/m²		149,874.19 W/m²		391,446.37 W/m²	
Diffusion Flame Height	1,843.75 μm		346.74 μm		1,612.03 μm		2,025.71 μm		724.44 μm	
Metal Combustion Heat Flux Density	461,056.91 W/m²		415,256.85 W/m²		529,304.99 W/m²		63,737.93 W/m²		150,917.35 W/m²	

3.44 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	32.23 mm/s		33.63 mm/s		23.12 mm/s		33.9 mm/s		12.38 mm/s	
Calculated Burning Rate	30.11 mm/s		33.39 mm/s		22.25 mm/s		33.75 mm/s		12.4 mm/s	
Difference	2.12 mm/s		0.24 mm/s		0.87 mm/s		0.16 mm/s		-0.027 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	42.11 mm/s	14.06 mm/s	42.11 mm/s	19.15 mm/s	42.11 mm/s	6.96 mm/s	50.51 mm/s	15.61 mm/s	28.56 mm/s	5.48 mm/s
Surface Temperature	705.21 K	651.57 K	705.21 K	665.84 K	705.21 K	621.28 K	657.18 K	632.51 K	877.76 K	832.95 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.7e+07 W/m ²		1.7e+07 W/m ²		1.7e+07 W/m ²		1.44e+07 W/m ²		2.38e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	17.64 μm		17.64 μm		17.64 μm		21.21 μm		11.88 μm	
Kinetic Flame Heat Flux Density (Skeleton)	3.22e+06 W/m ²		1.19e+07 W/m ²		2.14e+06 W/m ²		3.45e+06 W/m ²		133,479.12 W/m ²	
Kinetic Flame Height (Skeleton)	29.21 μm		16.47 μm		28.98 μm		27.76 μm		567.92 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.54e+06 W/m ²		2.62e+06 W/m ²		111,891.24 W/m ²		3.16e+06 W/m ²		9.71e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	48.34 μm		64.88 μm		376.01 μm		54.77 μm		15.78 μm	
Diffusion Flame Heat Flux Density	142,802.16 W/m ²		778,695.88 W/m ²		162,653.19 W/m ²		132,155.72 W/m ²		351,408.96 W/m ²	
Diffusion Flame Height	2,067.49 μm		386.31 μm		1,819.03 μm		2,295.39 μm		806.2 μm	
Metal Combustion Heat Flux Density	551,680.44 W/m ²		491,785.57 W/m ²		639,831.47 W/m ²		77,611.78 W/m ²		178,577.47 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	33.53 mm/s		37.52 mm/s		25 mm/s		37.87 mm/s		13.67 mm/s	
Difference	1.57 mm/s		-0.18 mm/s		0.92 mm/s		0.14 mm/s		-0.038 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	50.15 mm/s	15.5 mm/s	50.15 mm/s	21.1 mm/s	50.15 mm/s	7.76 mm/s	60.84 mm/s	17.35 mm/s	34.71 mm/s	5.99 mm/s
Surface Temperature	714.58 K	656.03 K	714.58 K	670.46 K	714.58 K	625.8 K	661.27 K	634.66 K	883.38 K	835.26 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.14e+07 W/m ²		2.14e+07 W/m ²		2.14e+07 W/m ²		1.79e+07 W/m ²		2.94e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	13.96 μm		13.96 μm		13.96 μm		16.98 μm		9.6 μm	
Kinetic Flame Heat Flux Density (Skeleton)	4.15e+06 W/m ²		1.53e+07 W/m ²		2.73e+06 W/m ²		4.38e+06 W/m ²		128,756.16 W/m ²	
Kinetic Flame Height (Skeleton)	22.53 μm		12.75 μm		22.51 μm		21.83 μm		586.96 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.68e+06 W/m ²		2.72e+06 W/m ²		115,470.95 W/m ²		3.26e+06 W/m ²		1.02e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	46.41 μm		62.23 μm		360.43 μm		53.12 μm		15.05 μm	
Diffusion Flame Heat Flux Density	129,275.11 W/m ²		705,327.33 W/m ²		145,586.97 W/m ²		118,768.33 W/m ²		321,099.81 W/m ²	
Diffusion Flame Height	2,280.38 μm		425.84 μm		2,029.16 μm		2,552.31 μm		881.58 μm	
Metal Combustion Heat Flux Density	643,212.93 W/m ²		572,770.7 W/m ²		759,568 W/m ²		91,735.44 W/m ²		205,550.9 W/m ²	

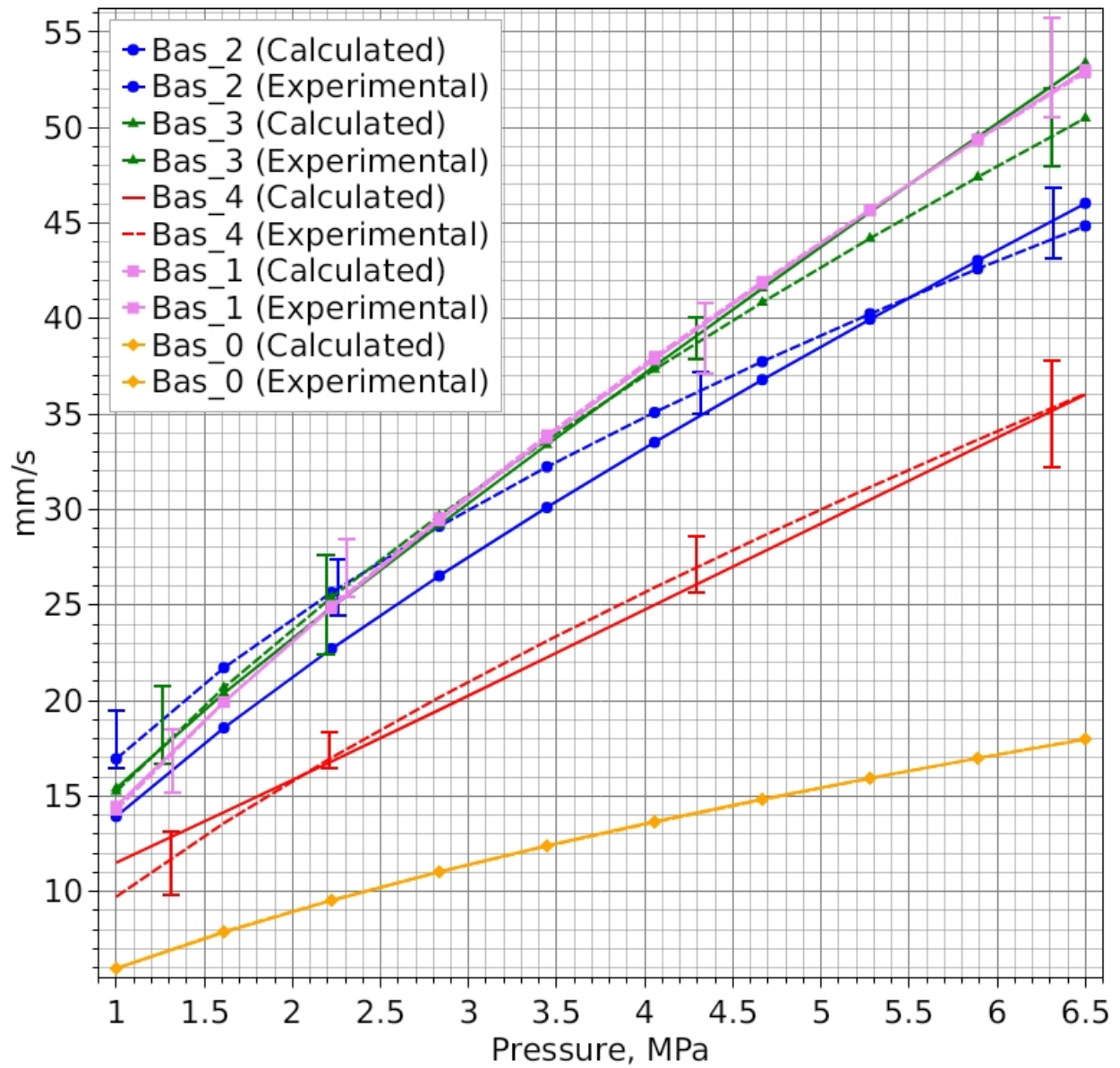
4.67 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	37.75 mm/s		40.85 mm/s		28.6 mm/s		41.93 mm/s		14.8 mm/s	
Calculated Burning Rate	36.8 mm/s		41.58 mm/s		27.76 mm/s		41.83 mm/s		14.84 mm/s	
Difference	0.95 mm/s		-0.74 mm/s		0.83 mm/s		0.11 mm/s		-0.037 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	58.31 mm/s	16.89 mm/s	58.31 mm/s	23.05 mm/s	58.31 mm/s	8.57 mm/s	71.41 mm/s	19.03 mm/s	41.04 mm/s	6.47 mm/s
Surface Temperature	722.86 K	659.98 K	722.86 K	674.71 K	722.86 K	629.96 K	664.83 K	636.54 K	888.26 K	837.24 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.61e+07 W/m ²		2.61e+07 W/m ²		2.61e+07 W/m ²		2.17e+07 W/m ²		3.53e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	11.42 μm		11.42 μm		11.42 μm		14.03 μm		7.99 μm	
Kinetic Flame Heat Flux Density (Skeleton)	5.16e+06 W/m ²		1.9e+07 W/m ²		3.36e+06 W/m ²		5.37e+06 W/m ²		124,871.61 W/m ²	
Kinetic Flame Height (Skeleton)	18.05 μm		10.27 μm		18.2 μm		17.77 μm		603.63 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.8e+06 W/m ²		2.8e+06 W/m ²		118,045.34 W/m ²		3.34e+06 W/m ²		1.06e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	44.87 μm		60.3 μm		349.05 μm		51.81 μm		14.45 μm	
Diffusion Flame Heat Flux Density	118,483.57 W/m ²		644,900.53 W/m ²		131,657.37 W/m ²		108,227.27 W/m ²		297,251.03 W/m ²	
Diffusion Flame Height	2,484.75 μm		465.08 μm		2,240.69 μm		2,799.16 μm		951.64 μm	
Metal Combustion Heat Flux Density	735,733.73 W/m ²		657,461.34 W/m ²		887,419.79 W/m ²		106,101.71 W/m ²		231,839.11 W/m ²	

5.28 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	40.24 mm/s		44.2 mm/s		31.17 mm/s		45.71 mm/s		15.92 mm/s	
Calculated Burning Rate	39.96 mm/s		45.58 mm/s		30.52 mm/s		45.66 mm/s		15.94 mm/s	
Difference	0.28 mm/s		-1.38 mm/s		0.65 mm/s		0.046 mm/s		-0.023 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	66.57 mm/s	18.23 mm/s	66.57 mm/s	24.97 mm/s	66.57 mm/s	9.38 mm/s	82.19 mm/s	20.65 mm/s	47.53 mm/s	6.92 mm/s
Surface Temperature	730.3 K	663.55 K	730.3 K	678.6 K	730.3 K	633.8 K	667.98 K	638.22 K	892.58 K	838.98 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	3.11e+07 W/m ²		3.11e+07 W/m ²		3.11e+07 W/m ²		2.56e+07 W/m ²		4.14e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	9.58 μm		9.58 μm		9.58 μm		11.87 μm		6.8 μm	
Kinetic Flame Heat Flux Density (Skeleton)	6.24e+06 W/m ²		2.28e+07 W/m ²		4.01e+06 W/m ²		6.41e+06 W/m ²		121,609.55 W/m ²	
Kinetic Flame Height (Skeleton)	14.87 μm		8.52 μm		15.16 μm		14.85 μm		618.39 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.9e+06 W/m ²		2.87e+06 W/m ²		119,947.03 W/m ²		3.4e+06 W/m ²		1.09e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	43.61 μm		58.79 μm		340.31 μm		50.73 μm		13.94 μm	
Diffusion Flame Heat Flux Density	109,628.34 W/m ²		594,599.04 W/m ²		120,121.14 W/m ²		99,668.85 W/m ²		277,916.6 W/m ²	
Diffusion Flame Height	2,682.2 μm		503.77 μm		2,452.69 μm		3,037.84 μm		1,017.22 μm	
Metal Combustion Heat Flux Density	829,295.24 W/m ²		744,980.53 W/m ²		1.02e+06 W/m ²		120,705.89 W/m ²		257,473.72 W/m ²	

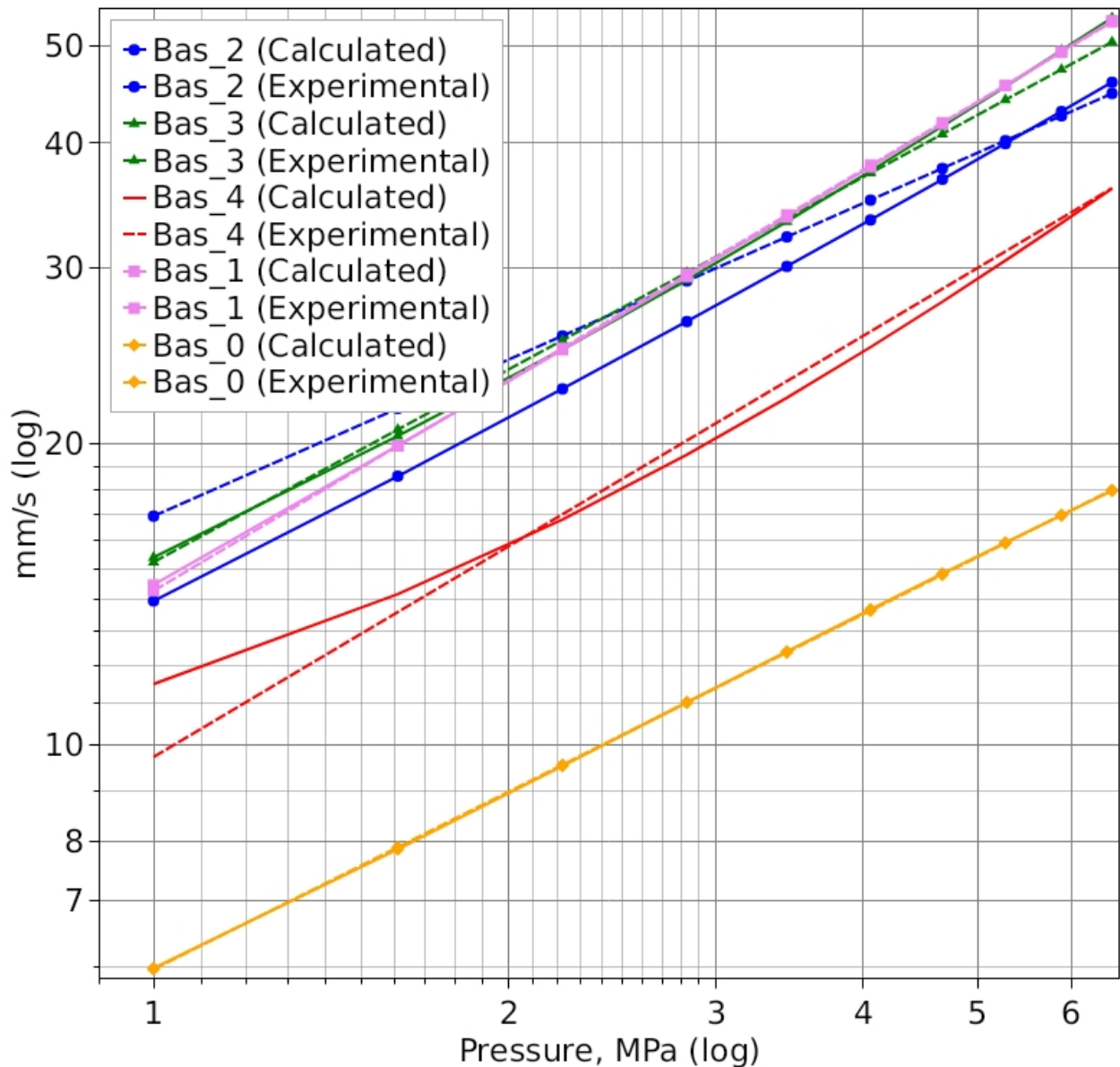
5.89 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	42.6 mm/s		47.41 mm/s		33.65 mm/s		49.35 mm/s		16.98 mm/s	
Calculated Burning Rate	43.04 mm/s		49.51 mm/s		33.27 mm/s		49.39 mm/s		16.98 mm/s	
Difference	-0.44 mm/s		-2.11 mm/s		0.38 mm/s		-0.038 mm/s		0.004 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	74.93 mm/s	19.54 mm/s	74.93 mm/s	26.86 mm/s	74.93 mm/s	10.19 mm/s	93.17 mm/s	22.23 mm/s	54.17 mm/s	7.34 mm/s
Surface Temperature	737.07 K	666.8 K	737.07 K	682.19 K	737.07 K	637.37 K	670.82 K	639.74 K	896.47 K	840.52 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	3.62e+07 W/m ²		3.62e+07 W/m ²		3.62e+07 W/m ²		2.97e+07 W/m ²		4.77e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	8.2 μm		8.2 μm		8.2 μm		10.23 μm		5.9 μm	
Kinetic Flame Heat Flux Density (Skeleton)	7.38e+06 W/m ²		2.68e+07 W/m ²		4.68e+06 W/m ²		7.51e+06 W/m ²		118,824.67 W/m ²	
Kinetic Flame Height (Skeleton)	12.53 μm		7.23 μm		12.9 μm		12.67 μm		631.58 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3.99e+06 W/m ²		2.92e+06 W/m ²		121,379.56 W/m ²		3.46e+06 W/m ²		1.13e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	42.56 μm		57.58 μm		333.36 μm		49.82 μm		13.49 μm	
Diffusion Flame Heat Flux Density	102,201.66 W/m ²		552,105.62 W/m ²		110,431.69 W/m ²		92,553.81 W/m ²		261,874.11 W/m ²	
Diffusion Flame Height	2,873.93 μm		541.89 μm		2,664.67 μm		3,269.73 μm		1,078.94 μm	
Metal Combustion Heat Flux Density	923,929.4 W/m ²		834,983.81 W/m ²		1.16e+06 W/m ²		135,544.1 W/m ²		282,484.81 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	46.04 mm/s		53.38 mm/s		36.02 mm/s		53.03 mm/s		17.96 mm/s	
Difference	-1.19 mm/s		-2.88 mm/s		0.044 mm/s		-0.14 mm/s		0.041 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	83.37 mm/s	20.81 mm/s	83.37 mm/s	28.72 mm/s	83.37 mm/s	11 mm/s	104.33 mm/s	23.77 mm/s	60.95 mm/s	7.73 mm/s
Surface Temperature	743.3 K	669.79 K	743.3 K	685.52 K	743.3 K	640.69 K	673.4 K	641.13 K	900 K	841.91 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	4.16e+07 W/m ²		4.16e+07 W/m ²		4.16e+07 W/m ²		3.39e+07 W/m ²		5.42e+07 W/m ²	
Kinetic Flame Height (Inter-Pocket)	7.13 μm		7.13 μm		7.13 μm		8.95 μm		5.18 μm	
Kinetic Flame Heat Flux Density (Skeleton)	8.57e+06 W/m ²		3.1e+07 W/m ²		5.37e+06 W/m ²		8.65e+06 W/m ²		116,410.87 W/m ²	
Kinetic Flame Height (Skeleton)	10.75 μm		6.24 μm		11.18 μm		10.98 μm		643.49 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	4.07e+06 W/m ²		2.97e+06 W/m ²		122,476 W/m ²		3.52e+06 W/m ²		1.16e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	41.67 μm		56.58 μm		327.67 μm		49.04 μm		13.1 μm	
Diffusion Flame Heat Flux Density	95,863.91 W/m ²		515,789.4 W/m ²		102,190.53 W/m ²		86,526.7 W/m ²		248,304.38 W/m ²	
Diffusion Flame Height	3,060.81 μm		579.4 μm		2,876.31 μm		3,495.88 μm		1,137.35 μm	
Metal Combustion Heat Flux Density	1.02e+06 W/m ²		927,069.23 W/m ²		1.31e+06 W/m ²		150,612.3 W/m ²		306,919.04 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Burning Rates (log-log) - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341E-05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

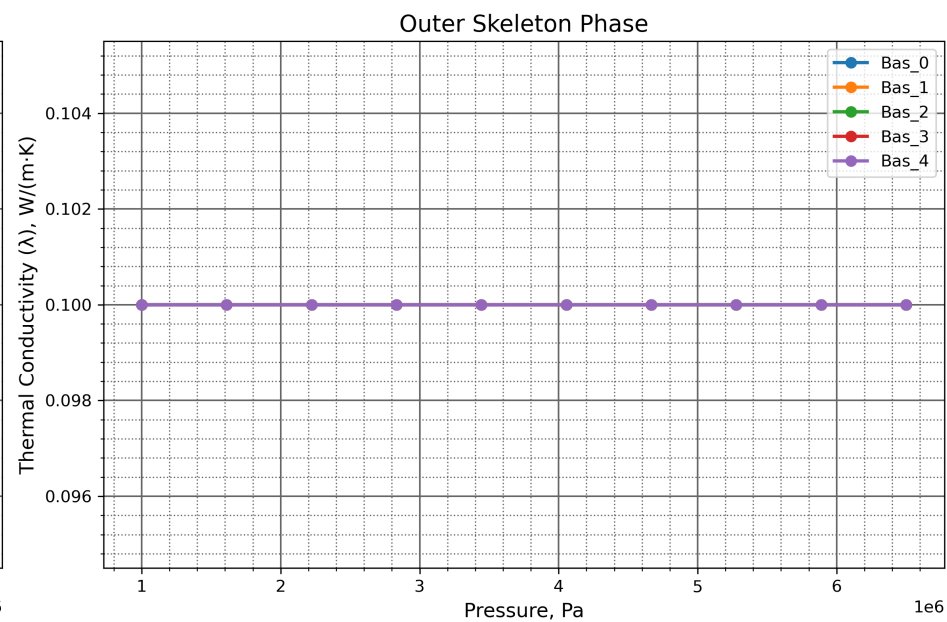
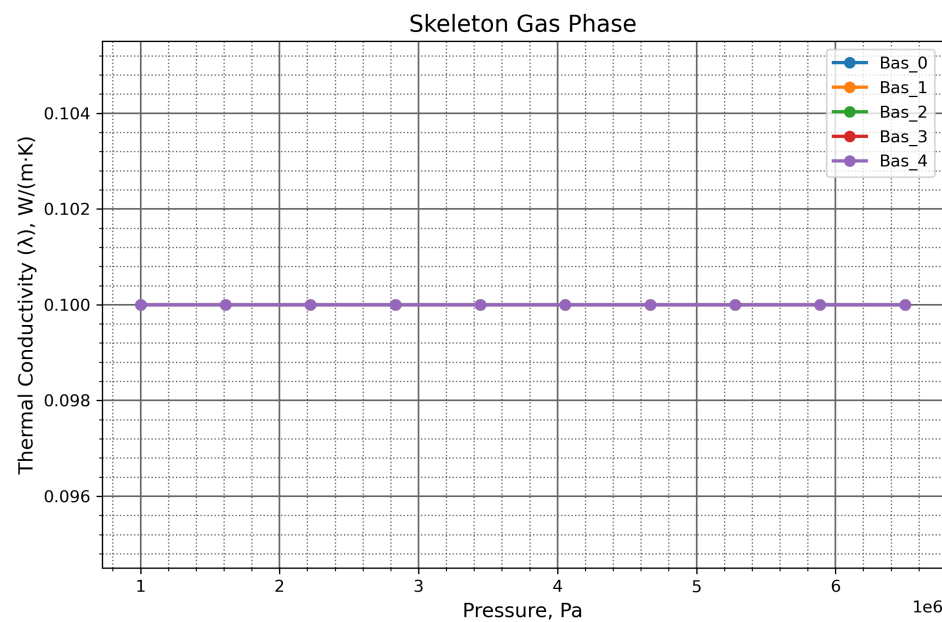
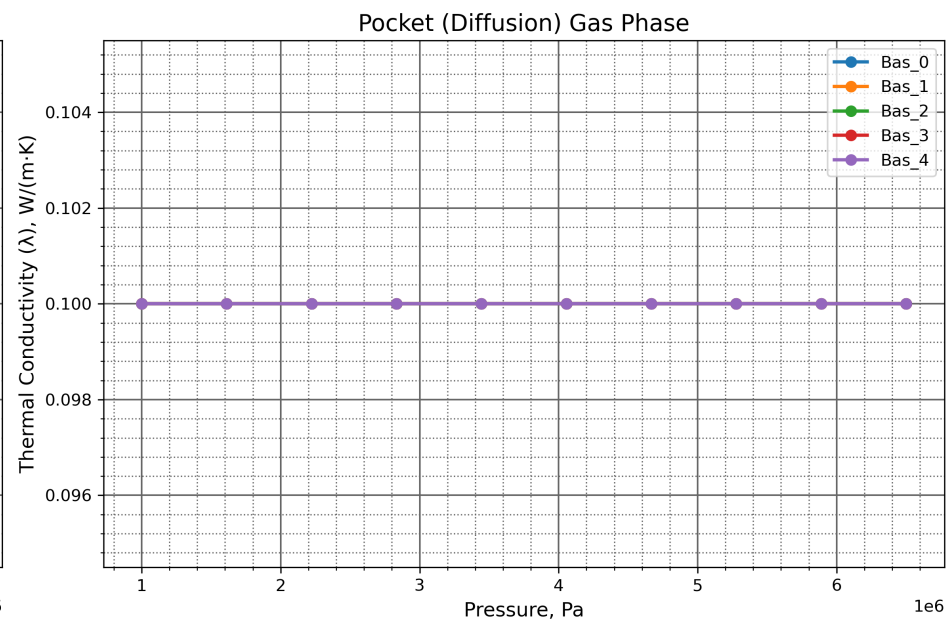
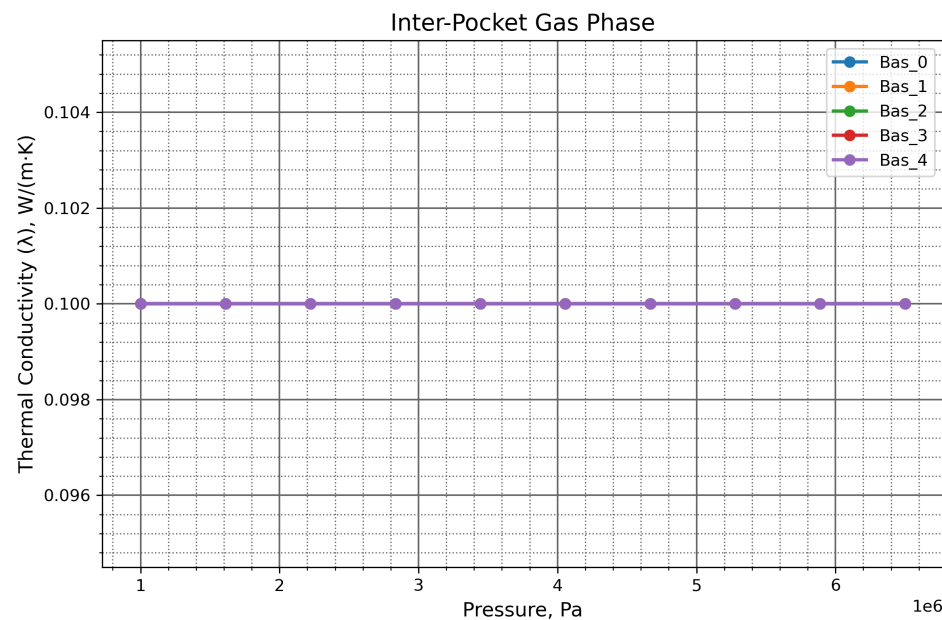
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

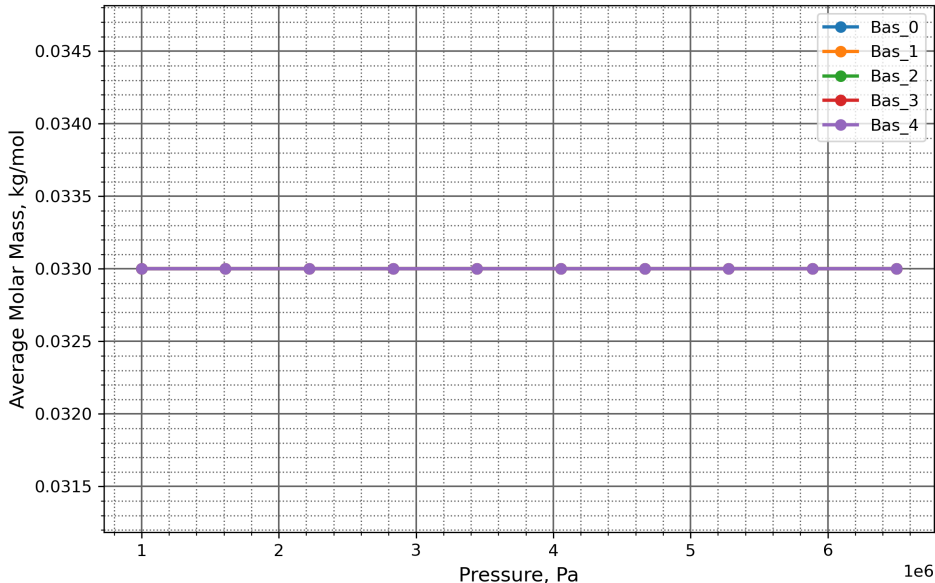
Kinetic flame temperature 2,365 K

Thermal Conductivity (λ), W/(m·K)

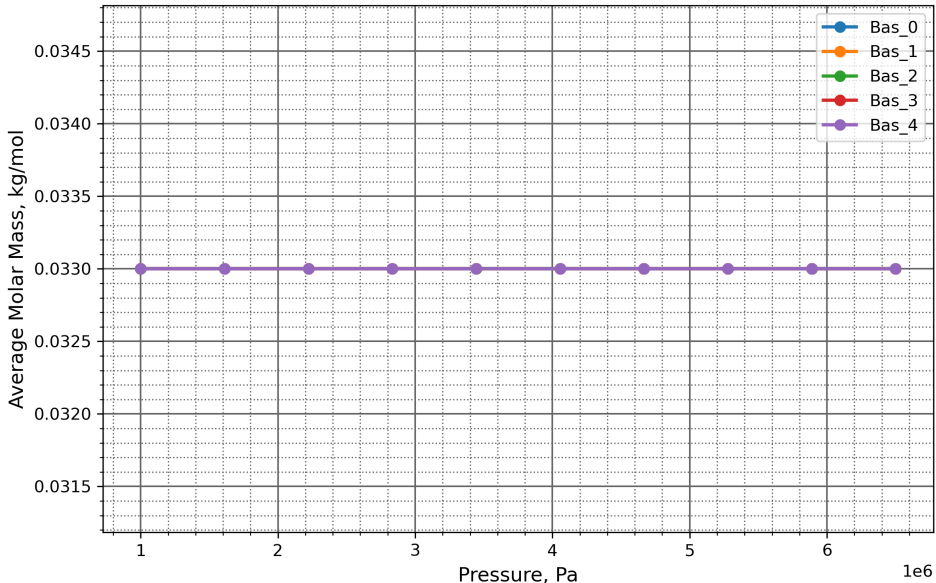


Average Molar Mass, kg/mol

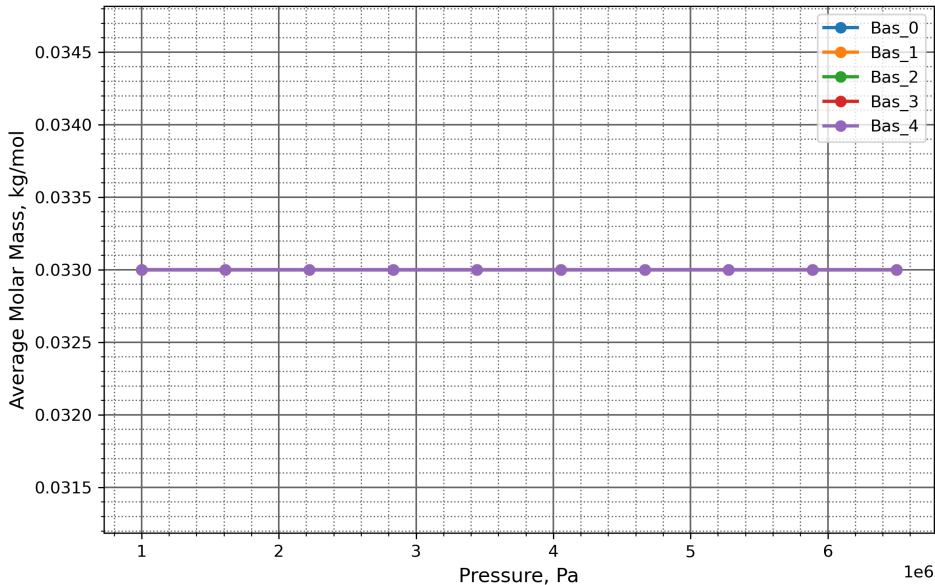
Inter-Pocket Gas Phase



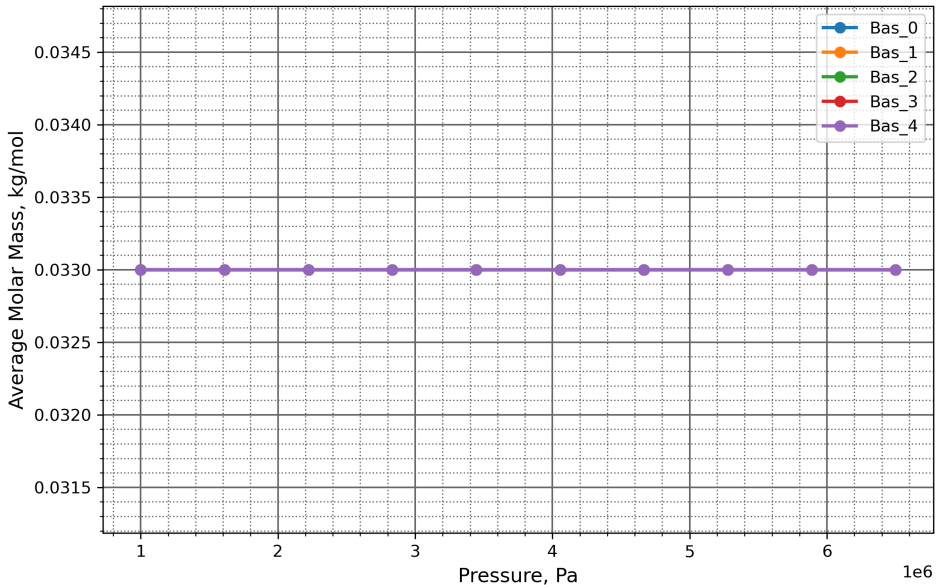
Pocket (Diffusion) Gas Phase



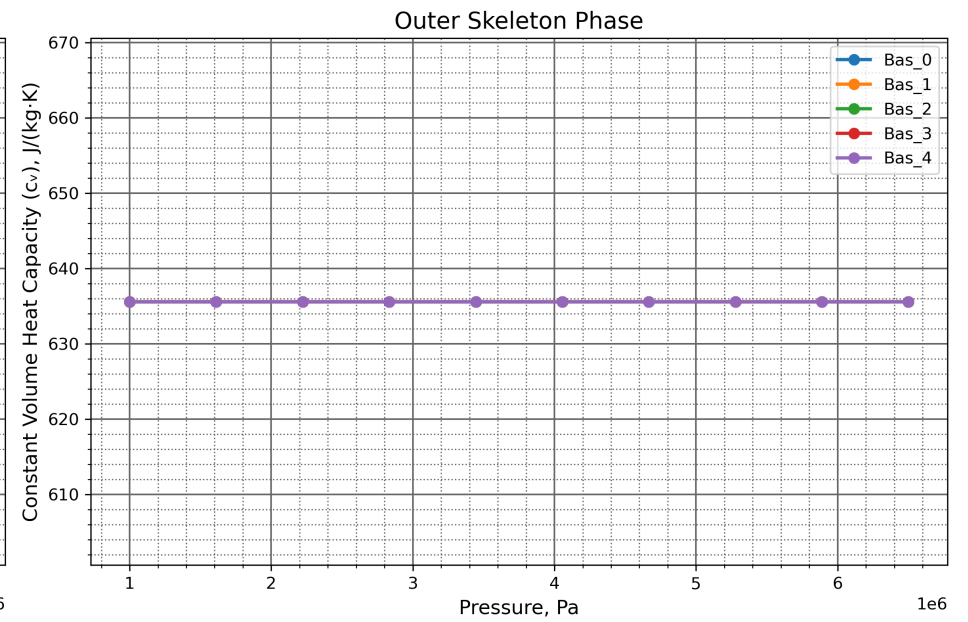
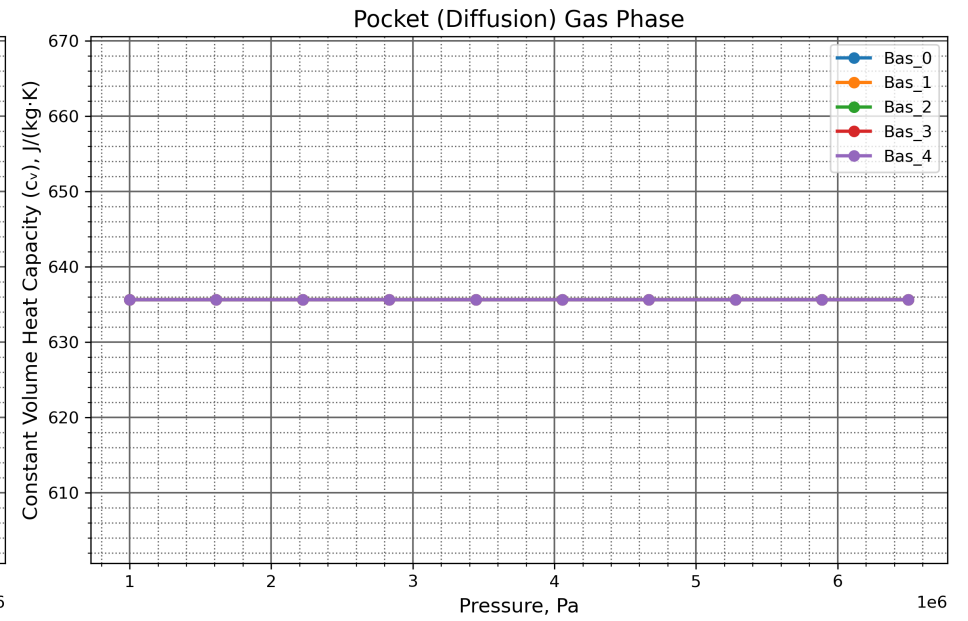
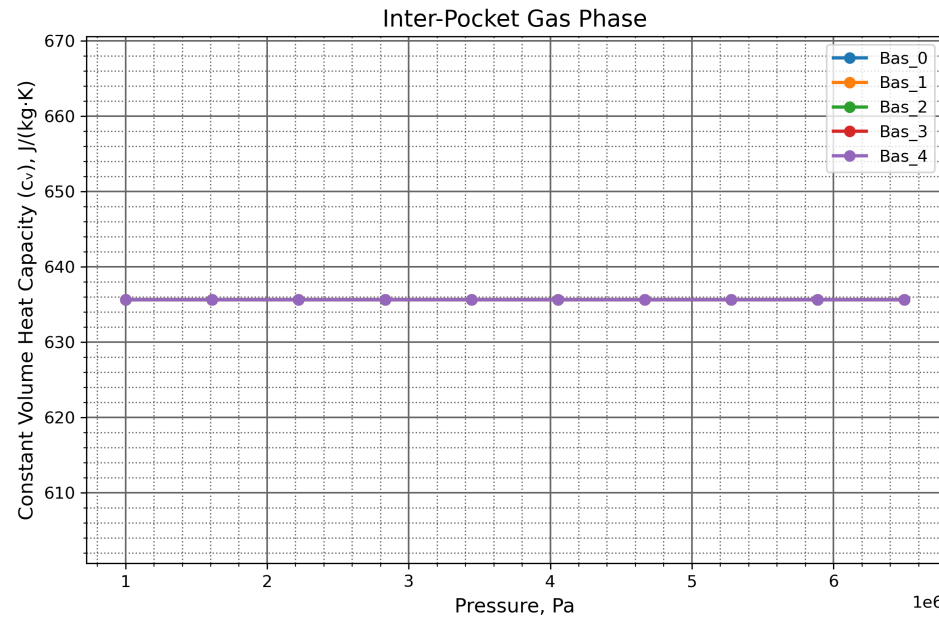
Skeleton Gas Phase



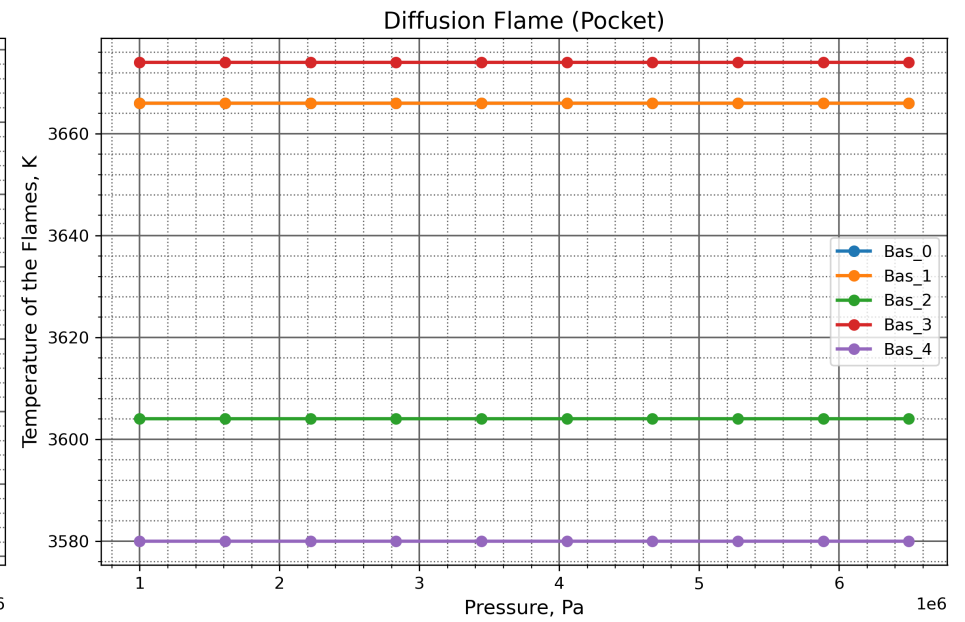
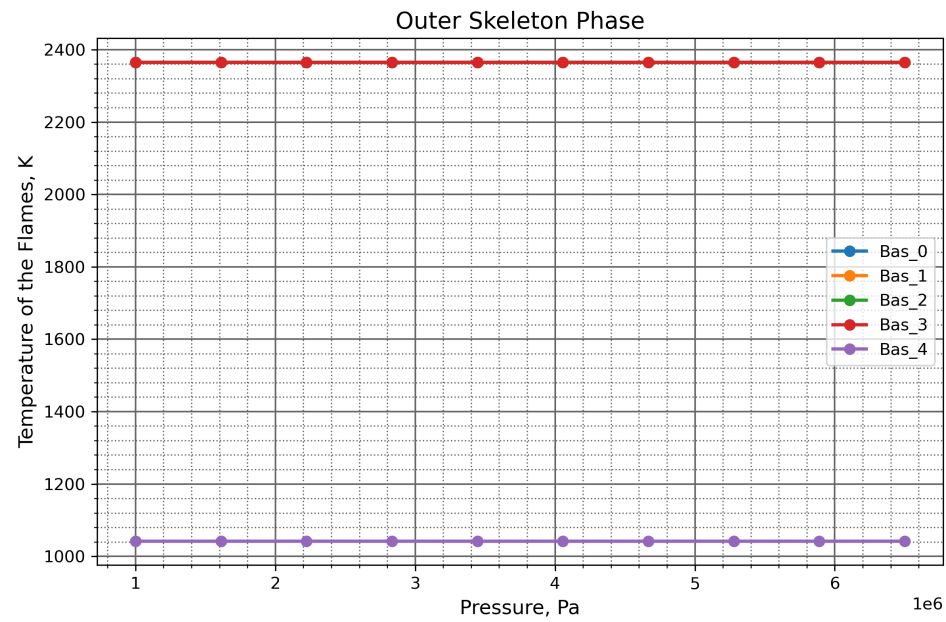
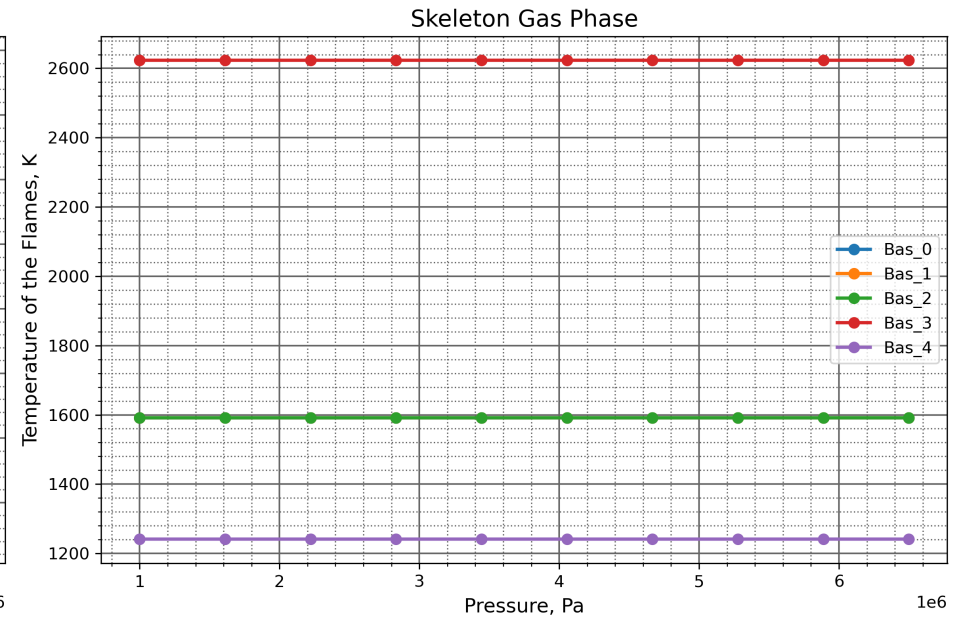
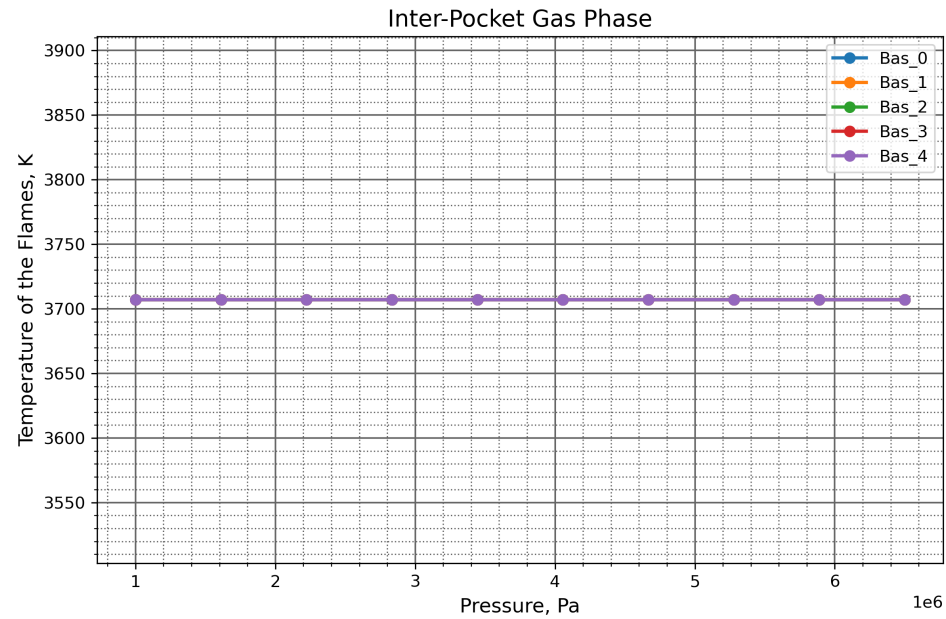
Outer Skeleton Phase



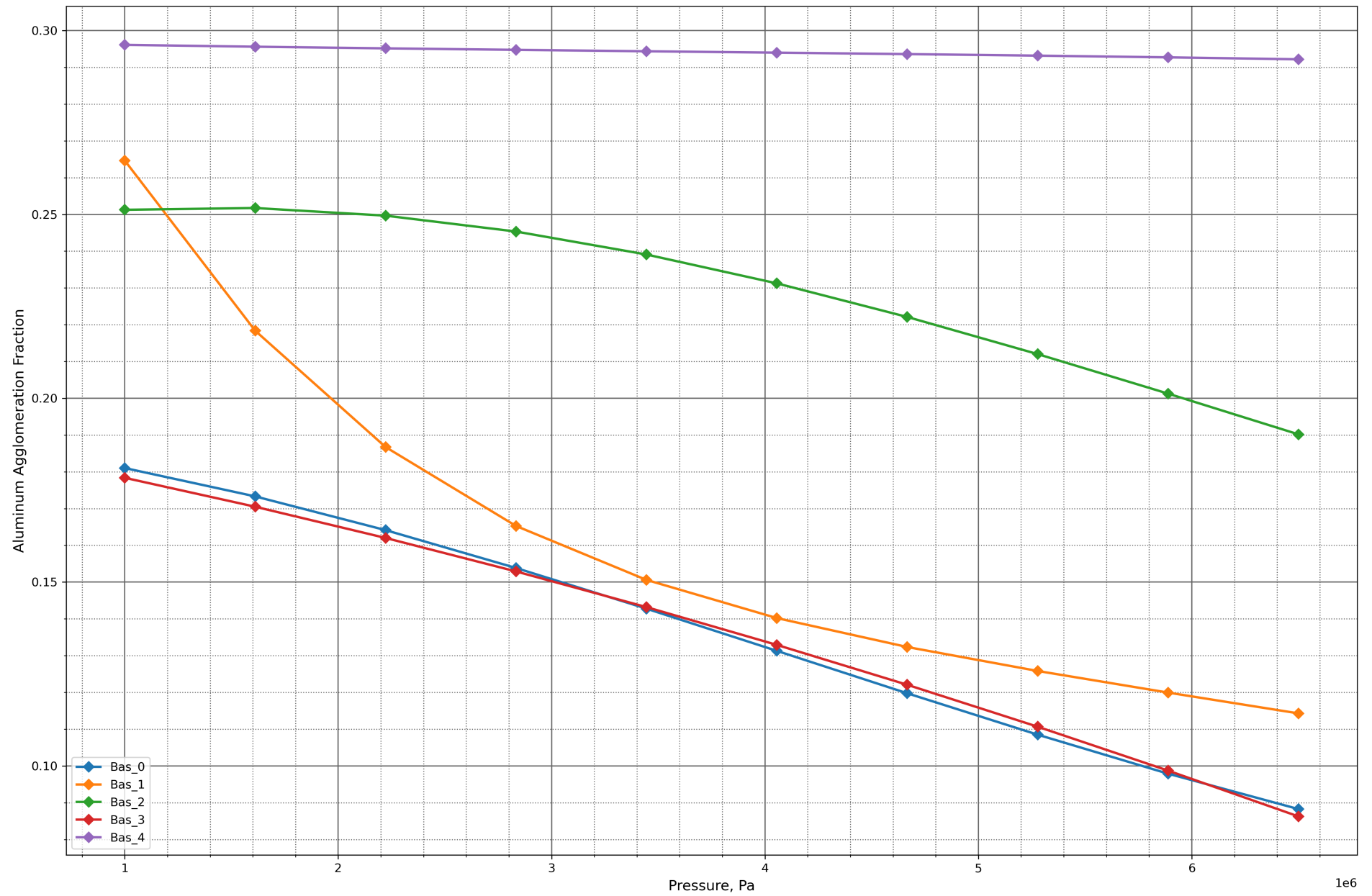
Constant Volume Heat Capacity (c_v), J/(kg·K)



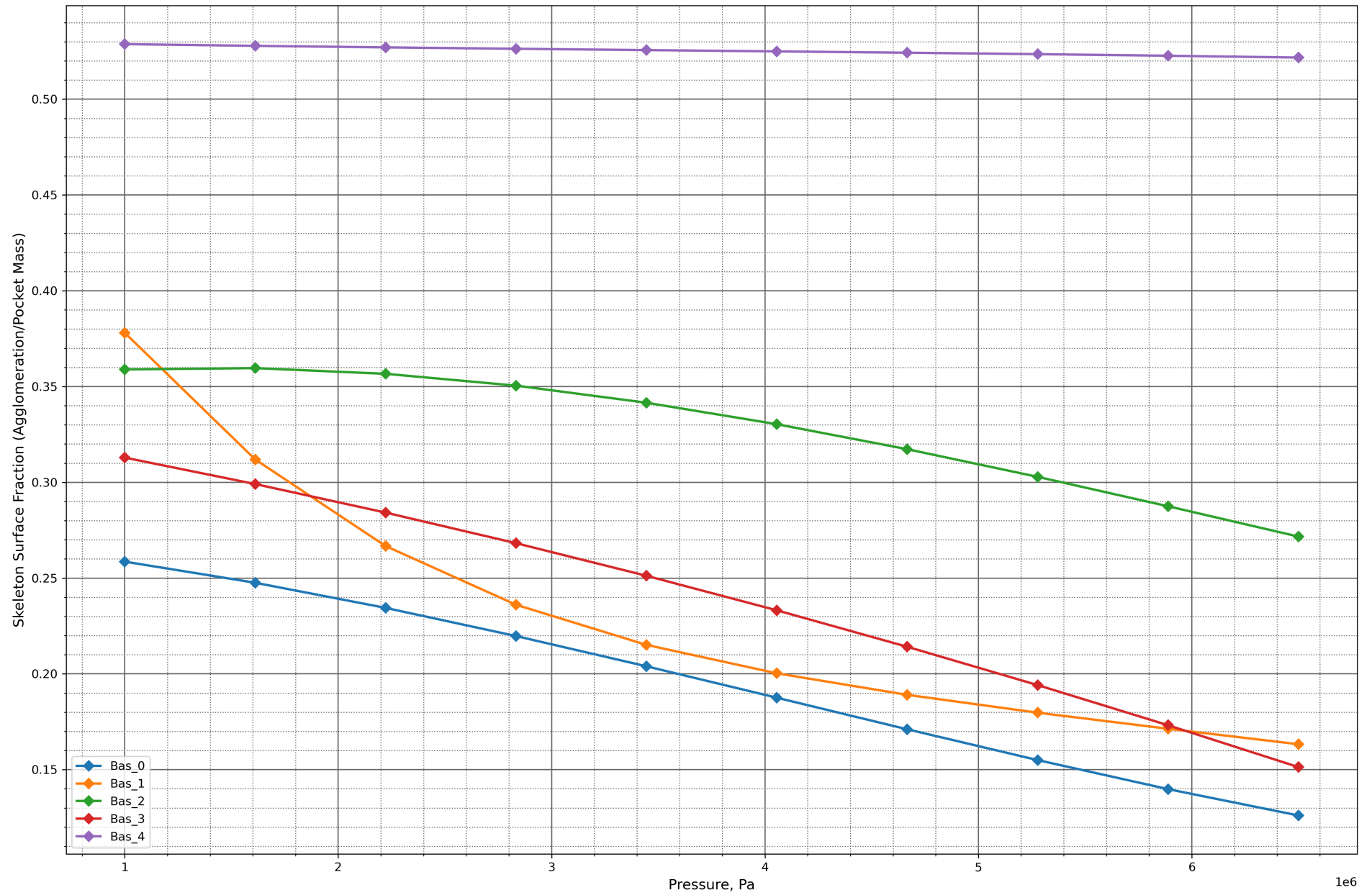
Temperature of the Flames, K



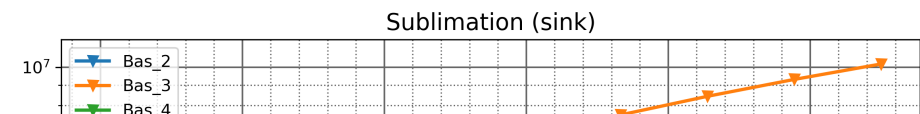
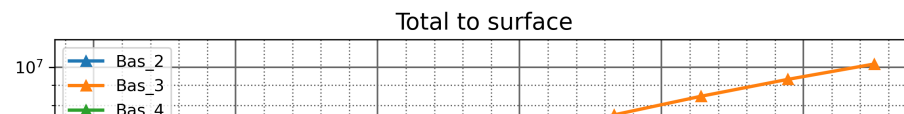
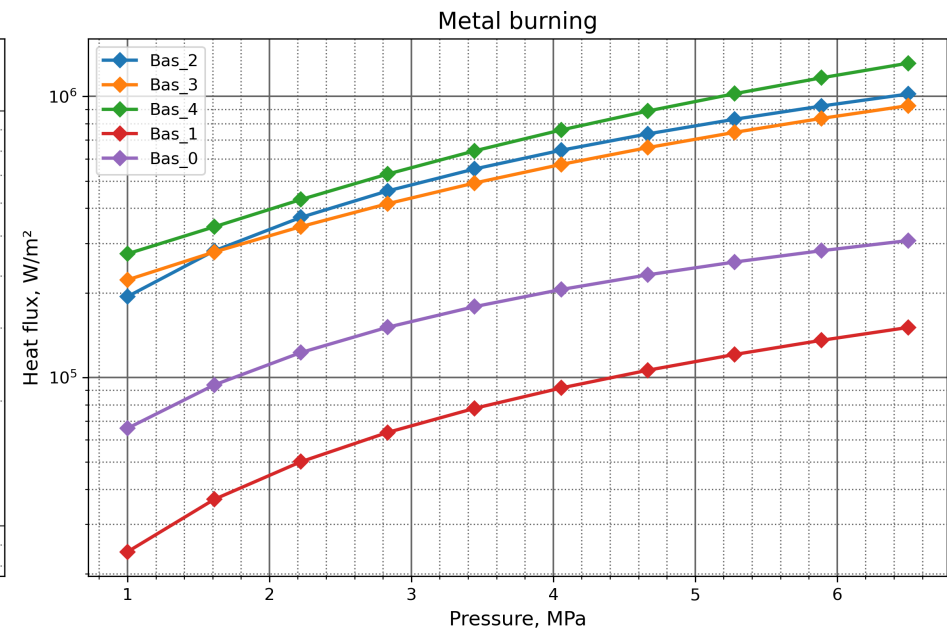
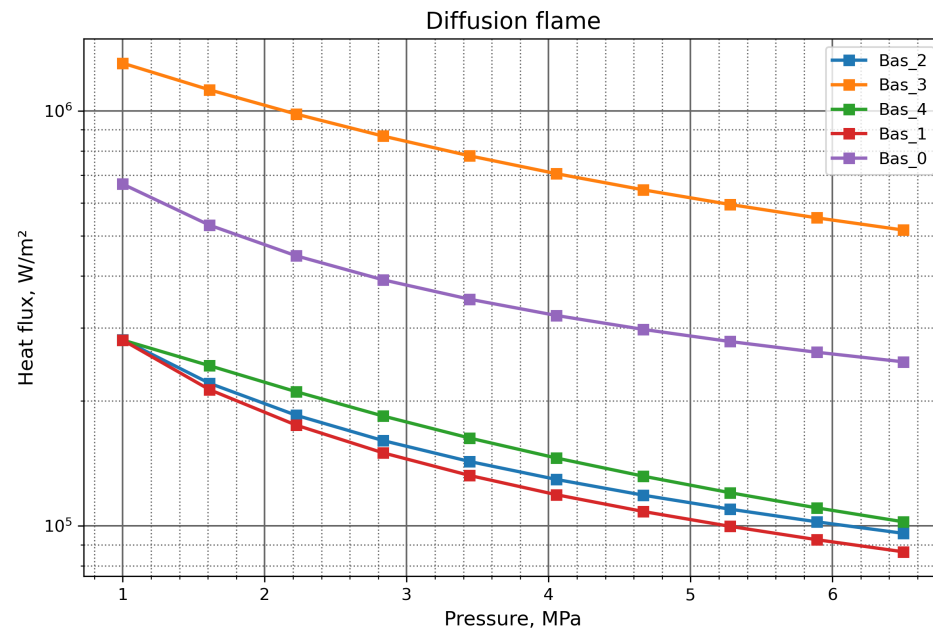
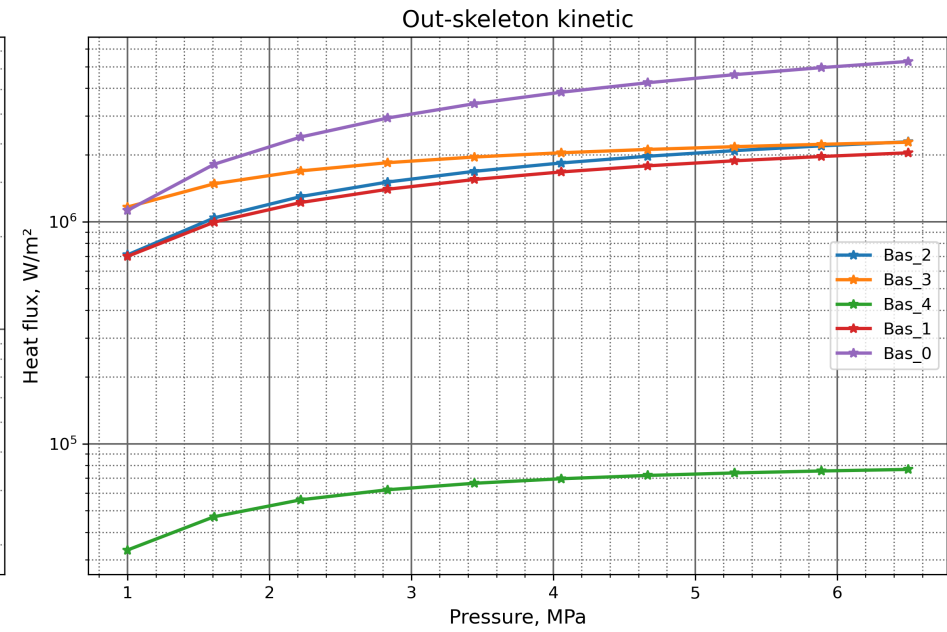
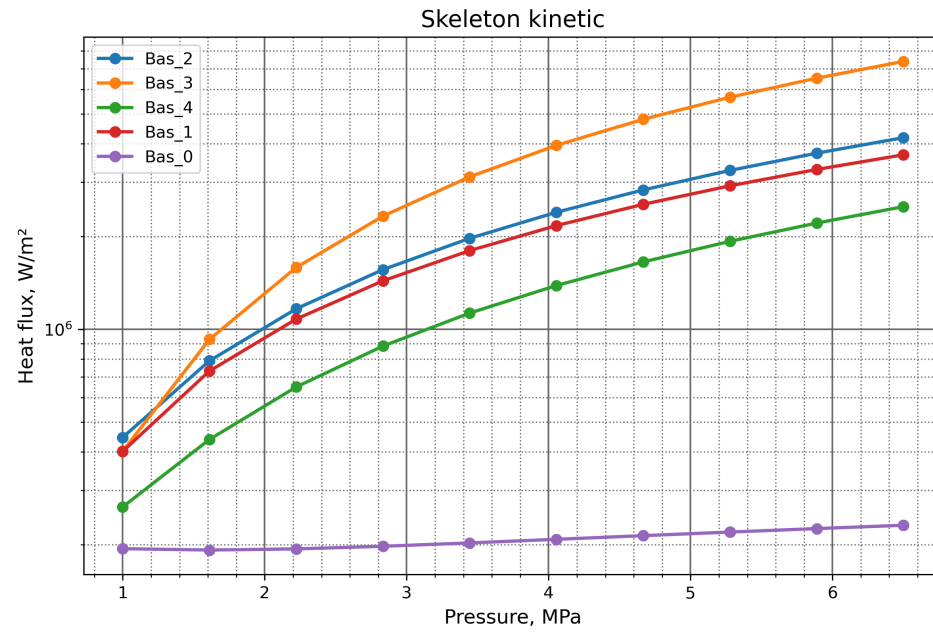
Aluminum Agglomeration Fraction



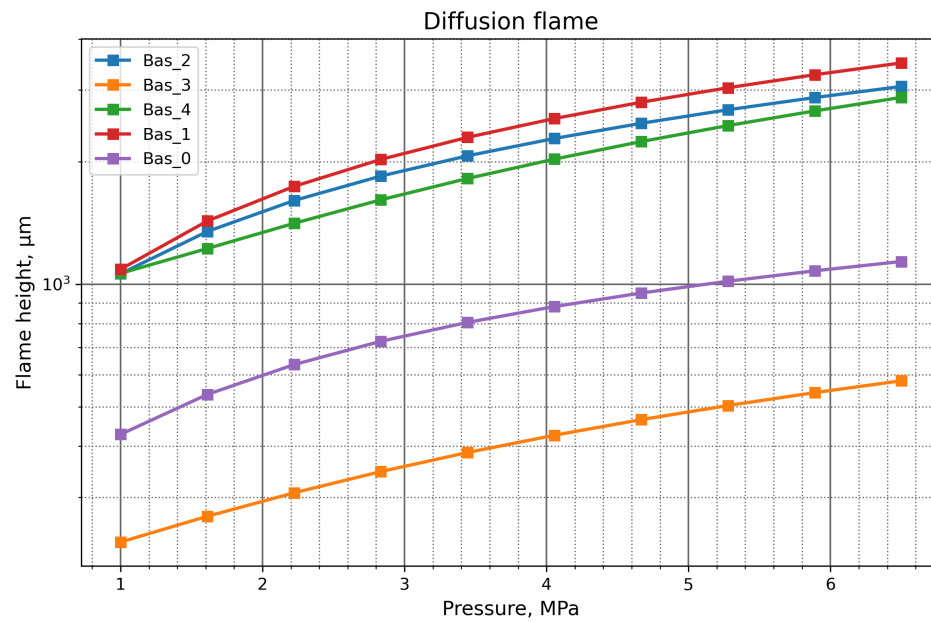
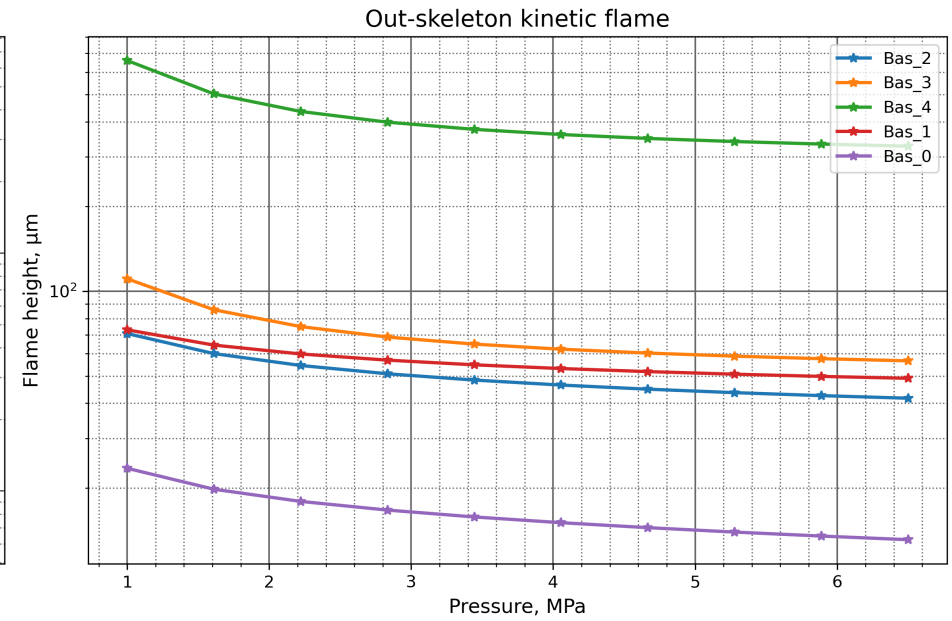
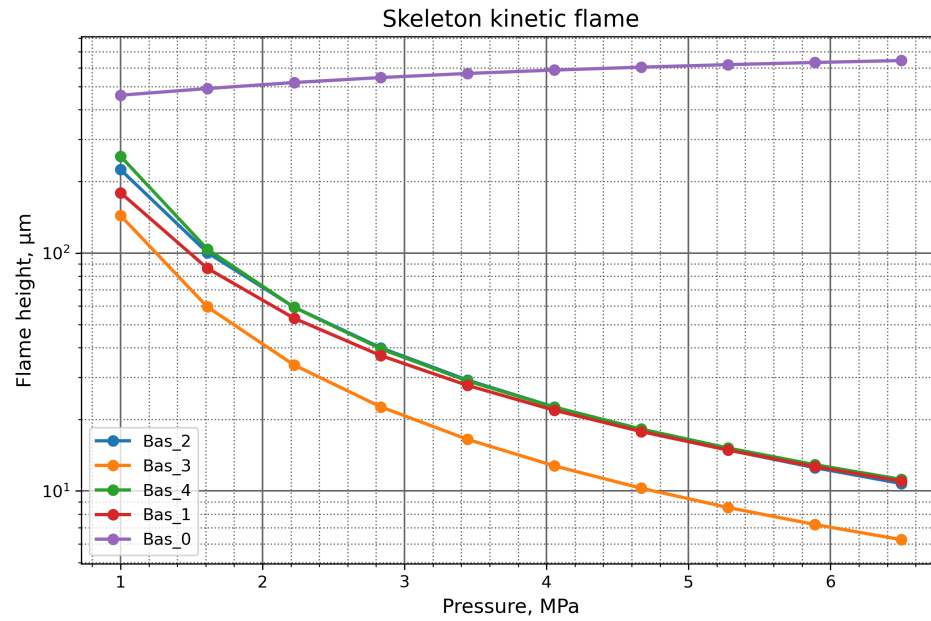
Skeleton Surface Fraction (Agglomeration/Pocket Mass)



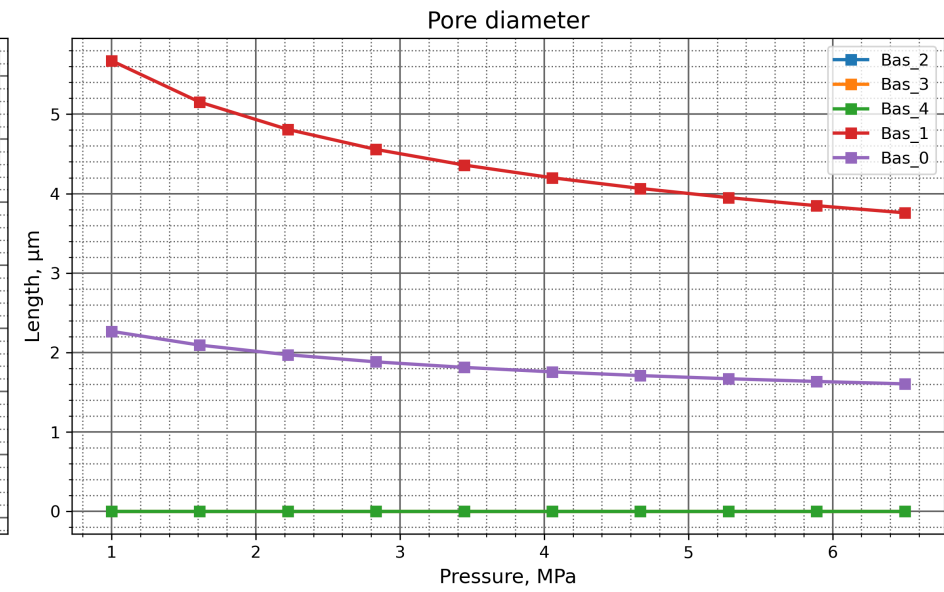
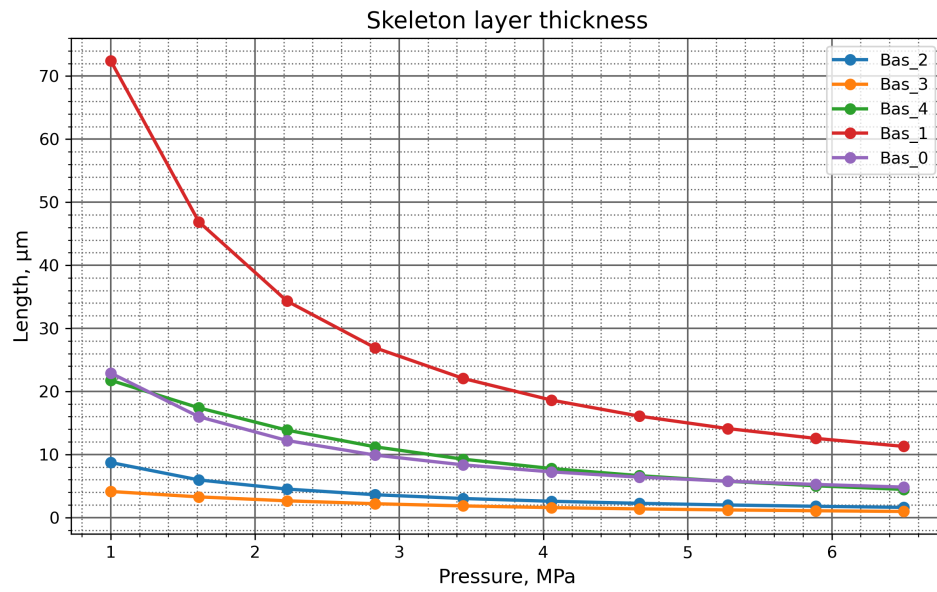
Pocket Heat-Flux Decomposition



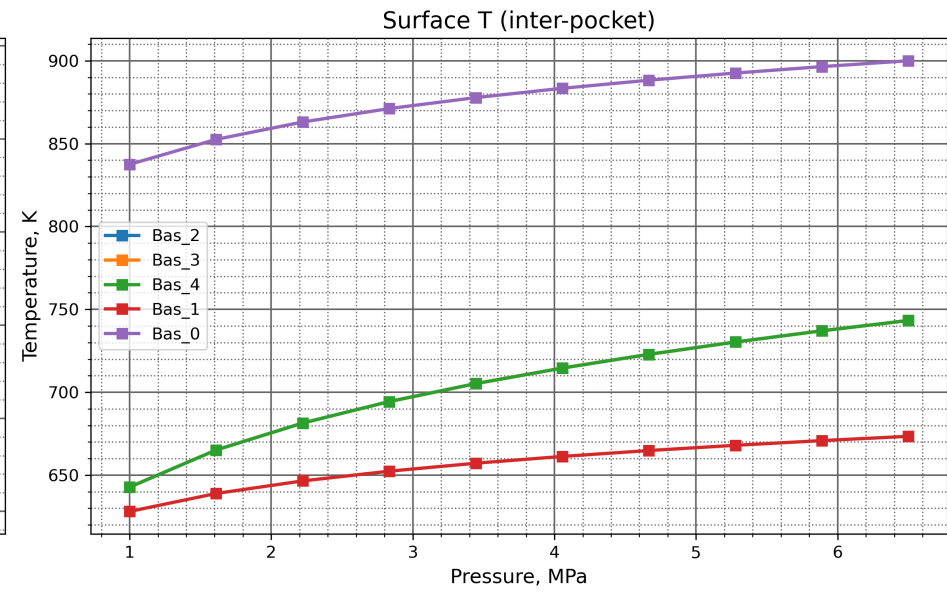
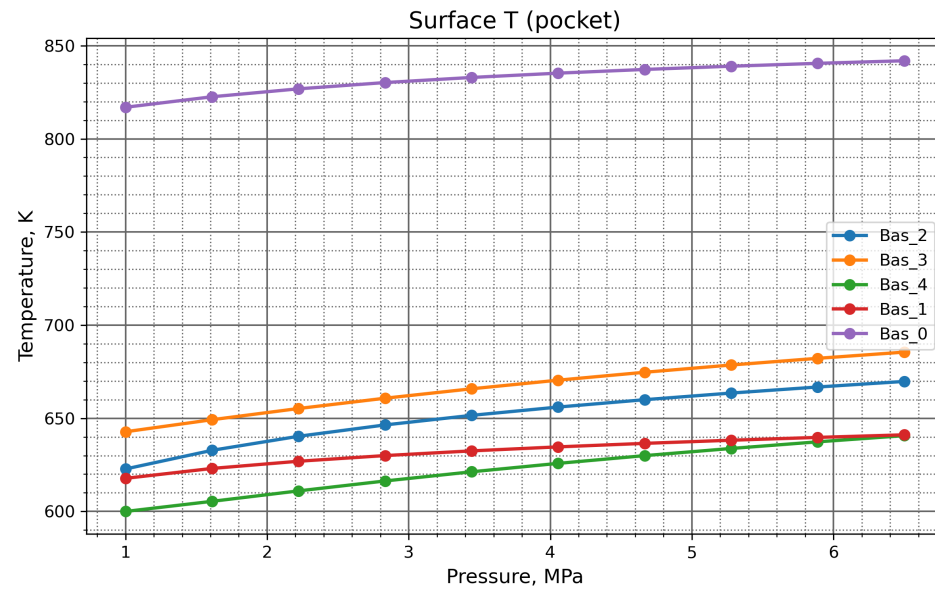
Pocket Flame Heights



Skeleton-Layer Geometry



Pocket Exit Temperatures



Performance Report

System Information

Operating System: Unix 6.8.0.134 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.10 (.NET 10.0.10)
Architecture: X64 / X64

Total Execution Time

Total execution time: 06:01:18.085

Garbage Collection Information

Total Memory: 10480048 bytes
GC Pause Time Percentage: 0.13%
Generation 0 Collections: 23250
Generation 1 Collections: 2036
Generation 2 Collections: 215

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 15324235
Maximum execution time: 30.8288 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.7900214460427525 ms
Execution time standard deviation: 0.17051141022451372 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 10997374
Maximum execution time: 17.1159 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2381055848696514 ms
Execution time standard deviation: 0.11648809111220354 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 9027895
Maximum execution time: 15.5716 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.28224829330644075 ms
Execution time standard deviation: 0.048903886793958955 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 15223836
Maximum execution time: 32.6959 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7883712762342134 ms
Execution time standard deviation: 0.16818357166855163 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 10897424
Maximum execution time: 18.9052 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2374935753165226 ms
Execution time standard deviation: 0.11633894259469342 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 8926105
Maximum execution time: 21.2758 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.2821686478480945 ms
Execution time standard deviation: 0.04675668272709031 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 50.8676 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.7900189965787895 ms
Execution time standard deviation: 0.16840201949407843 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 10900623
Maximum execution time: 14.6218 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23752049465428868 ms
Execution time standard deviation: 0.1159434363690469 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 8931193
Maximum execution time: 18.6658 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28180410479315904 ms
Execution time standard deviation: 0.048096535464218555 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 29.9401 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.7866363390147294 ms
Execution time standard deviation: 0.1666877035567668 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 10897628
Maximum execution time: 17.24 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.23714595826726778 ms
Execution time standard deviation: 0.11553138210500799 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 8929321
Maximum execution time: 15.307 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2817184158683375 ms
Execution time standard deviation: 0.046241215113830406 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 34.9967 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.795202933373816 ms
Execution time standard deviation: 0.16812345303903556 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 10901859
Maximum execution time: 19.8032 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2389220754919067 ms
Execution time standard deviation: 0.11694154047233668 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 8929154

Maximum execution time: 19.5836 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28385590135413524 ms

Execution time standard deviation: 0.04624745252288816 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 23.5013 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7890024774505254 ms

Execution time standard deviation: 0.16654802777689054 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 10900389

Maximum execution time: 25.7616 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23821399706011812 ms

Execution time standard deviation: 0.11659983180708329 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 8932086

Maximum execution time: 19.7109 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28255123456043035 ms

Execution time standard deviation: 0.0464100568123121 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 28.9923 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7904445611737929 ms

Execution time standard deviation: 0.16603268904505647 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 10901146

Maximum execution time: 20.5566 ms

Minimum execution time: 0.0003 ms

Mean execution time: 0.2379491303116073 ms

Execution time standard deviation: 0.11589739518174573 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 8933165

Maximum execution time: 17.4513 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28246932434360894 ms

Execution time standard deviation: 0.04510855849527684 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 20.0873 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7914442793064608 ms

Execution time standard deviation: 0.16473774158873264 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 10899065

Maximum execution time: 14.9932 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2389467173743848 ms
Execution time standard deviation: 0.11592457672812996 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 8931570

Maximum execution time: 28.4994 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28405871525385146 ms
Execution time standard deviation: 0.04730069874575532 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 46.2349 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7854637514027246 ms
Execution time standard deviation: 0.1687351169462233 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 10900708

Maximum execution time: 23.3345 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23640822680508583 ms
Execution time standard deviation: 0.11593054251423035 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 8931184

Maximum execution time: 20.4961 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28073894102959396 ms
Execution time standard deviation: 0.0473589028642602 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 32.3991 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7912057444721211 ms
Execution time standard deviation: 0.16543047705153022 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 10900864

Maximum execution time: 24.3773 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.23879016324761365 ms
Execution time standard deviation: 0.11665111494547713 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 8931760

Maximum execution time: 23.6849 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28333314775588586 ms
Execution time standard deviation: 0.04610302038782499 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 24.6773 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7910371279090541 ms

Execution time standard deviation: 0.16401355983333332 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 10895344

Maximum execution time: 19.3206 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2385762272122827 ms

Execution time standard deviation: 0.11558564258270784 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 8928735

Maximum execution time: 32.441 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2836362760458187 ms

Execution time standard deviation: 0.046810304810507575 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 29.1694 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.792321069722495 ms

Execution time standard deviation: 0.16897674146669633 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 10897106

Maximum execution time: 20.794 ms

Minimum execution time: 0.0003 ms

Mean execution time: 0.23819040915085998 ms

Execution time standard deviation: 0.11648446441053432 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 8925159

Maximum execution time: 18.5563 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2828848079793595 ms

Execution time standard deviation: 0.046322985083106644 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 36.8633 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7918337080746821 ms

Execution time standard deviation: 0.165184430490542 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 10896698

Maximum execution time: 24.0477 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23828097822843736 ms

Execution time standard deviation: 0.1165646916070798 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 8923773

Maximum execution time: 25.4208 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2829779724227033 ms

Execution time standard deviation: 0.045174247217534776 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 15223836
Maximum execution time: 31.1401 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7901949920376572 ms
Execution time standard deviation: 0.16675587506788692 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 10896092
Maximum execution time: 14.3474 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23738402651153342 ms
Execution time standard deviation: 0.11521987652418901 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 8925241
Maximum execution time: 22.5717 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2822864209045012 ms
Execution time standard deviation: 0.04712466674036703 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 15223836
Maximum execution time: 26.981 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7833860252960405 ms
Execution time standard deviation: 0.16858034254307405 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 10897477
Maximum execution time: 21.2445 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23602485574413773 ms
Execution time standard deviation: 0.11624790054870283 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 8929357
Maximum execution time: 24.2629 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2806850687120753 ms
Execution time standard deviation: 0.048021228298979195 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 15223836
Maximum execution time: 26.4283 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7912410652873422 ms
Execution time standard deviation: 0.1654914364623798 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 10899410
Maximum execution time: 27.8069 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.237955468974942 ms
Execution time standard deviation: 0.11584372100447843 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 8931066
Maximum execution time: 16.0771 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.2827931497987367 ms
Execution time standard deviation: 0.045424620926214875 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 37.8995 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.789823428333051 ms
Execution time standard deviation: 0.16559067079145848 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 10901348
Maximum execution time: 21.7778 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2377585604367658 ms
Execution time standard deviation: 0.1156251195772919 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 8932398
Maximum execution time: 21.3458 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28431272501516913 ms
Execution time standard deviation: 0.045626380847060805 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 28.0096 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.7909980021067143 ms
Execution time standard deviation: 0.16517184332596718 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 10902494
Maximum execution time: 12.0171 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.2381253167578067 ms
Execution time standard deviation: 0.11530359639263854 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 8933313
Maximum execution time: 21.1333 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2833540819066428 ms
Execution time standard deviation: 0.04541573196771669 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 24.0712 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7938814514423851 ms
Execution time standard deviation: 0.16697048501979692 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 10901130
Maximum execution time: 30.9178 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24018298500247512 ms
Execution time standard deviation: 0.1172246501700351 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 8932275
Maximum execution time: 14.5881 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28492519192479476 ms
Execution time standard deviation: 0.045274152405899916 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 21.1726 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7904895035587812 ms
Execution time standard deviation: 0.16472642052432798 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 10902451
Maximum execution time: 21.747 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.23813532842293383 ms
Execution time standard deviation: 0.11615722850918743 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 8929255
Maximum execution time: 20.3004 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2826927991416987 ms
Execution time standard deviation: 0.04521166136086911 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 43.9601 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7908689084473245 ms
Execution time standard deviation: 0.16800658398016008 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 10899610
Maximum execution time: 15.2714 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23814828370006827 ms
Execution time standard deviation: 0.11637405099823421 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 8926972
Maximum execution time: 14.9246 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28218752872757524 ms
Execution time standard deviation: 0.04628546105883162 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 23.0981 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.7895365918418473 ms
Execution time standard deviation: 0.16493708238599403 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 10897630
Maximum execution time: 24.3432 ms

Minimum execution time: 0.0003 ms
Mean execution time: 0.2378637380788084 ms
Execution time standard deviation: 0.11578422955194712 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 8929950
Maximum execution time: 18.7906 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2826624656016788 ms
Execution time standard deviation: 0.045880877034442144 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 25.469 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7894890946013492 ms
Execution time standard deviation: 0.1662483253942651 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 10899370
Maximum execution time: 17.6316 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2375230880225006 ms
Execution time standard deviation: 0.11555294838247947 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 8932384
Maximum execution time: 14.3374 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2822389272113764 ms
Execution time standard deviation: 0.045264207838802445 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 30.758 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7882212831312372 ms
Execution time standard deviation: 0.16750224415299647 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 10896937
Maximum execution time: 25.196 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23768239017073783 ms
Execution time standard deviation: 0.11627020823372862 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 8930015
Maximum execution time: 17.8749 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.2822672555309382 ms
Execution time standard deviation: 0.04577535241578497 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 15223836
Maximum execution time: 49.0841 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7883991553443067 ms
Execution time standard deviation: 0.16693933106927622 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 10895100

Maximum execution time: 19.7705 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23716257735128768 ms

Execution time standard deviation: 0.11518292802426368 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 8927491

Maximum execution time: 14.8327 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2819787004881952 ms

Execution time standard deviation: 0.04510507920760391 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 27.5471 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7880318501724185 ms

Execution time standard deviation: 0.16835344536093455 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 10898942

Maximum execution time: 19.7914 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23756655753377995 ms

Execution time standard deviation: 0.11559859279976398 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 8928870

Maximum execution time: 15.0791 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.283285151155721 ms

Execution time standard deviation: 0.046342047608025794 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 31.1268 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7896309302333046 ms

Execution time standard deviation: 0.16606309217254167 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 10899107

Maximum execution time: 15.8953 ms

Minimum execution time: 0.0003 ms

Mean execution time: 0.2382387113182484 ms

Execution time standard deviation: 0.11582902686063262 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 8927558

Maximum execution time: 29.4942 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2825597228491797 ms

Execution time standard deviation: 0.045859639771362586 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 25.0826 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7916657873614021 ms
Execution time standard deviation: 0.16467059794679476 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 10897870
Maximum execution time: 20.7785 ms
Minimum execution time: 0.0003 ms
Mean execution time: 0.23993462548186498 ms
Execution time standard deviation: 0.1167049649729568 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 8929010
Maximum execution time: 26.0371 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28349477080887653 ms
Execution time standard deviation: 0.04558753219965734 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 15223836
Maximum execution time: 27.7843 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7880283281494017 ms
Execution time standard deviation: 0.16655500835249676 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 10899809
Maximum execution time: 23.3802 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23705200463604104 ms
Execution time standard deviation: 0.11536747641968358 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 8928069
Maximum execution time: 19.8826 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2816539726675224 ms
Execution time standard deviation: 0.046285282671348 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 15223836
Maximum execution time: 20.0326 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7902171899907491 ms
Execution time standard deviation: 0.16446650430576104 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 10899419
Maximum execution time: 14.8761 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23816112668022746 ms
Execution time standard deviation: 0.11526646652629613 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 8932455
Maximum execution time: 15.5168 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2825279215848522 ms

Execution time standard deviation: 0.04297343767163315 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 22.2 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.79246744545848 ms

Execution time standard deviation: 0.162975916343261 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 10899173

Maximum execution time: 31.3783 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23893992752477888 ms

Execution time standard deviation: 0.11579353351242652 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 8934251

Maximum execution time: 11.8325 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2838433518824573 ms

Execution time standard deviation: 0.04190698242373994 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 15223836

Maximum execution time: 31.982 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7915261647458155 ms

Execution time standard deviation: 0.16653563678324462 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 10899156

Maximum execution time: 20.8952 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23790879600217382 ms

Execution time standard deviation: 0.11612454553103968 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 8927472

Maximum execution time: 18.2928 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28251689983456246 ms

Execution time standard deviation: 0.044351000092946374 ms